



CATALOGUE OF DEFINITIONS OFFICIAL MATCH DATA

Version 5.1 – (external)

DFL Deutsche Fußball Liga GmbH & SPORTEC SOLUTIONS GMBH



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1. INTRODUCTION

The present document provides the technical definitions and specifications of the official match data. Figure 1 gives an overview of the structure of the official match data. They are divided into three basic types, *master data*, *match data* and *positional data*, whereby

- *master data* represent data that are subject to few changes and are ordinarily available at the start of a season (e.g. *stadia*, *players*),
- *match data* refer to match information about the run-up to a match (e.g. *environmental conditions*, *teams*) or to match events during a match (e.g. *balls played*, *challenges*) and are collected by people, and
- *positional data* of the players and the ball are collected by a semi-automated optical tracking system based on image recognition (e.g. *positions* and *speeds*).

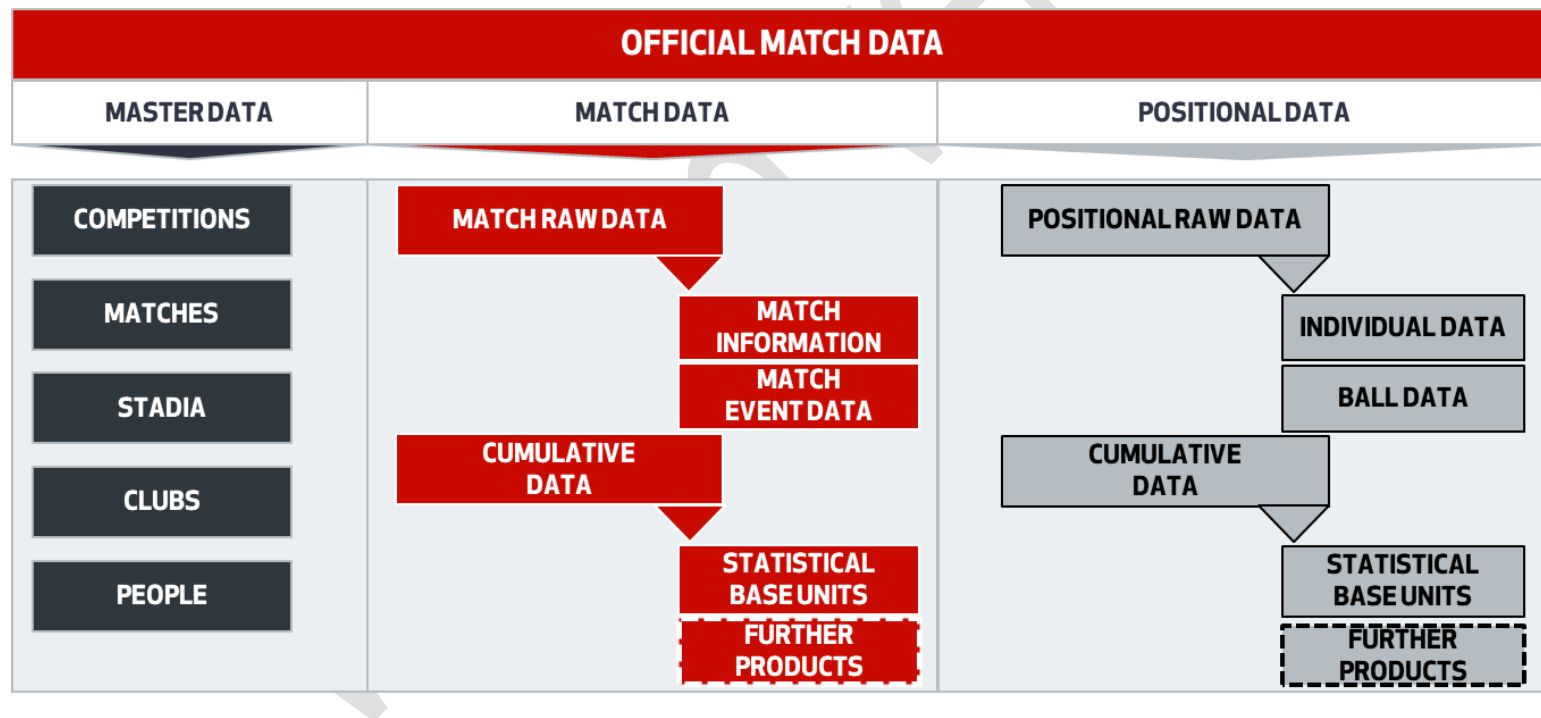


Figure 1: Structure of the official match data

Match data and positional data are both subdivided into *raw data* and *cumulative data*. *Raw data* include individual data packets (e.g. *match events* or *tracking frames*), which are combined during the course of a match to form cumulative data (e.g. *percentage of challenges won*, *distance covered*). In addition to providing technical definitions of the *raw data*, the aim of this catalogue of definitions is to establish a binding definition of the essential statistical *base units*, thus enabling uniform use in sports reporting and other data use processes. Other possible products based on game data are not specified in this document (e.g. *heat maps*).

Match data

Match raw data consist of *match information* and *match events* (see figure 2). *Match information* is usually available shortly before the start of a match and is subdivided into the categories *general*, *environment*, *teams*, *referees* and *other match information*. Match events are subdivided into the categories *player actions*, *team actions* and *referee actions*.

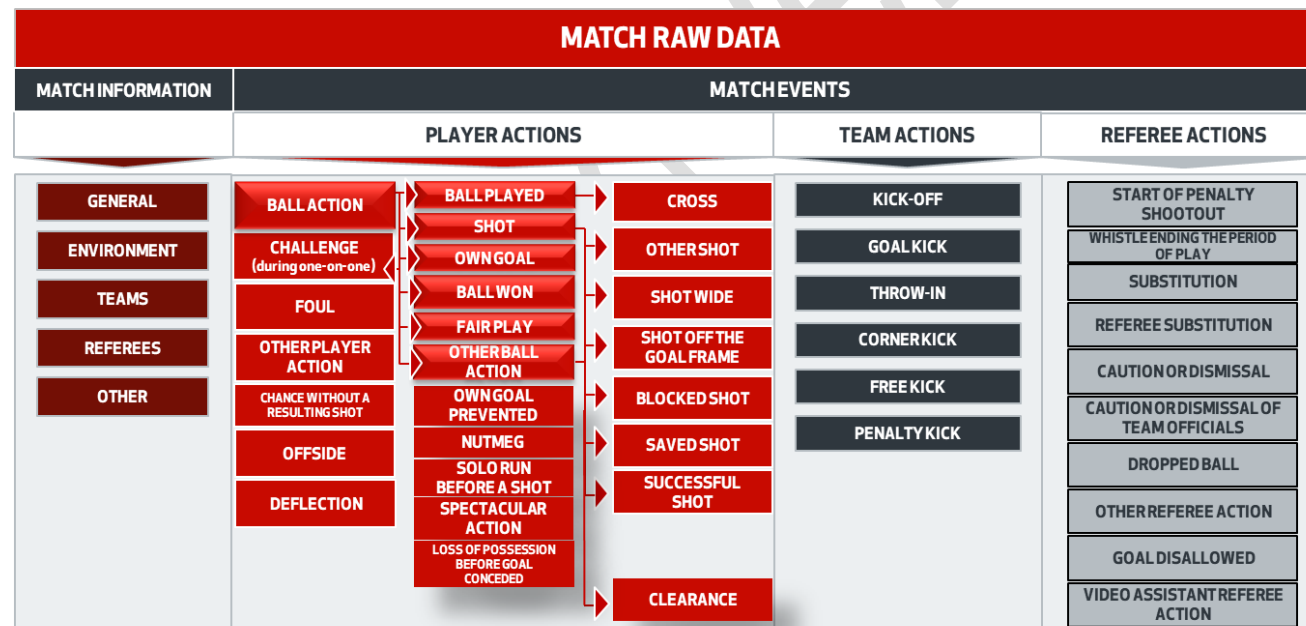


Figure 2: Structure of match raw data

Match event data

The central concept in the description of match events is their arrangement in a hierarchical order. The hierarchy is created by the use of a so-called specialisation relation (represented by the arrows in figure 2). A match event that is located below another event in the hierarchy is a special case of the latter (for example, a *ball played* is a special case of a *ball action*, which in turn is a special case of a *player action*). The attributes of an event are "inherited" by all subordinate events (for example, *player actions* and *team actions* contain a reference to a team and a player; this attribute is therefore also present in the subordinate events *ball action* and *ball played*).

Firstly, the definition of the individual match events takes place by assigning them to their distinct parent class and distinguishing them as precisely as possible from other match events of the same class (= siblings). For example, the definition of a *ball played*, assigned to the parent class *ball action*, is primarily used to distinguish it from its siblings (*fair play*, *throw-in*, etc.). Secondly, the attribute manifestations of a class are distinguished from one another. This is done either in reference to the DFB football rules (e.g. *direct/indirect* execution mode for *free kicks*) or by describing the characteristics as precisely as possible (e.g. *short*, *medium*, *long* for the event *pass* of the parent class *ball played*). Thirdly, rules of application, details and examples are presented, which arise as a result of the respective definition, and explanations are provided, which may refer to the conceptual basis of the definitions (= additional description).

Every match event contains the attribute time, which describes the time of occurrence during the match. These points in time reflect the order in which the match events occur within the game and therefore the actual sequence of events. For set pieces, the time attribute describes the moment of execution, not the moment in which the referee decided to award a particular set piece. The latter is described by the attribute decision time. The duration of a set piece situation is described by the difference between its end frame and its start frame. Additionally, all *player actions* and *team actions* contain x- and y-coordinates, which represent their respective positions on the pitch. If the location of a player's first touch of the ball (source position) deviates significantly (more than five metres) from the position of his subsequent ball action, both positions are recorded. In addition, if the ball leaves the pitch before a set piece and the set piece's position of execution deviates from the location where the ball left the pitch, two separate positions are recorded. For example, this is the case before a *corner kick*, a *goal kick* or before a *kick-off* following a *successful shot*.

Match event data are divided into player actions, team actions and referee actions.

Player actions are match events involving at least one player. These include *ball action*, *challenge*, *rule violation*, *other player action*, *chances without a resulting shot*, *offside and deflection*, if it is performed by a player. *Ball actions* include all actions of a player that require a touch of the ball. These are *ball played*, *shot*, *ball won*, *own goal*, *fair play* and *other ball action*. *Ball actions* are never collected without further specifications being made, just like shots. They are always recorded in their specialised form, which is then listed further down the hierarchical order (represented by shape and colour in figure 2).

Team actions are match events that are assigned to a team (also called set pieces). These include *kick-off*, *goal kick*, *throw-in*, *corner kick*, *free kick* and *penalty kick*. All set pieces include the attribute execution type, which contains a reference to a *ball action*. This *ball action* describes how the set piece is executed. This *ball action* – or more precisely the specialisation of the *ball action* – is collected with its attributes at a set piece (for example, if a *free kick* is executed as a *shot*, all attributes of both the event *free kick* as well as the event *shot* are supplied). A set piece is only recorded if it is executed.

Referee actions are *match events* during which the referee is the primary acting person. They include *start of penalty shootout*, *whistle ending the period of play*, *substitution*, *referee substitution*, *caution or dismissal*, *caution or dismissal of team officials*, *dropped ball*, *other referee action*, *goal disallowed* and *video assistant referee review*.

The concept of ball control and phases of possession

The concept of ball control is essential for the definition of *match events*. Ball control describes the possibility of implementing a tactical plan of action, which is based on perceptual processes. *Player actions* such as *ball played*, *ball won* or *shot* are examples of implementing a tactical plan of action. A team is deemed to have gained ball control as soon as one of its players has the possibility of implementing a tactical plan of action with the ball. Vice versa, a team's ball control is deemed to have ended once none of its players can implement a tactical plan of action with the ball and if at the same time the opposing team is able to do so. If a player can choose between different tactical actions and their implementation with the ball in a given match situation (e.g. *ball played* or *shot*), he is considered to have certain ball control. The distinction between ball control and certain ball control is made when a player has a certain amount of time to control the ball and to make a considered choice between his tactical plans of action.

Additionally, every *ball action* contains an attribute *phase of possession*. A *phase of possession* describes a period of time during which a single player has ball control. The period stretches from the player's first touch of the ball (e.g. a *ball won*) until ball control is lost (e.g. by a *ball played* or a *shot*). During one phase of possession a single player could execute one *ball action*, but also several different *ball actions* (e.g. a *ball won* followed by a *shot*). Therefore, the match consists of a sequence of *phases of possession*, which are numbered consecutively after being recorded. The attribute *phase of possession* indicates the *phase of possession* belonging to a certain *ball action*.

2. MASTER DATA

Competition		Information about the <i>competition</i> .
Competition ID	a) reference competition ID	<i>Competition ID</i> describes the continuous <i>Competition ID</i> set by the match data collector. The ID is assigned after consulting the DFL.
Official name	a) Bundesliga b) Bundesliga 2 c) relegation playoffs Bundesliga d) relegation playoffs Bundesliga 2 e) Supercup f) replays Bundesliga g) replays Bundesliga 2 h) replays relegation playoffs Bundesliga i) replays relegation playoffs Bundesliga 2 j) replays Supercup	<i>Official name</i> is the competition name assigned by the organiser. Replays within a competition are started or completed matches of the respective competition that were not counted, but that are played for a second time.
Country	a) Germany	<i>Country</i> specifies the country where the competition takes place.
Organiser	a) Die Liga - Fußballverband e.V. (league association)	<i>Organiser</i> is the organiser of the competition.
Sport	a) association football	Indicates the sport of the competition.
Type	a) league b) playoffs c) tournament	Type is the type of competition. The attribute tournament includes cup competitions.
Match		Information about a <i>match</i> .
Identification number	a) alphanumeric	<i>Identification number</i> is the unique match number assigned by the match data collector.

Official match number	a) alphanumeric	<i>Official match number</i> is the match number assigned by the organiser in reference to the stage of the season.
Competition	a) reference competition ID	<i>Competition</i> indicates the competition the match is played in.
Stadium	a) reference stadium ID	<i>Stadium</i> refers to the stadium in which the game takes place according to the official fixture list.
Home team	a) reference club ID	<i>Home team</i> refers to the home team assigned by the organiser.
Away team	a) reference club ID	<i>Away team</i> refers to the away team assigned by the organiser.
Season	a) 2017/2018 b) YYYY/YYYY	<i>Season</i> represents the season during which the game takes place.
Type	a) matchday first half of season b) matchday second half of season c) first leg d) second leg e) single game	<i>Type</i> is the type of match. For league matchdays a distinction is to be made between the first and second half of the season. For playoff games a distinction is to be made between first leg and second leg. Single game is used for the Supercup game.
Matchday	a) numeric	<i>Matchday</i> indicates the continuously numbered matchday of the season (1-34) on which the game takes place.
Date and kick-off time set	a) yes b) no	Describes whether the match date and kick-off time of a match have been set.
Planned date and kick-off time	a) alphanumeric	Planned date and kick-off time is the planned date and time when the match is to start in the UTC format.
Characterisation of date and kick-off time	a) cancelled b) postponed c) counted d) stopped e) replayed f) continued g) annulled	<i>Characterisation of date and kick-off time</i> describes the status of the planned date and time of kick-off. Cancelled is used when the game has not yet begun, no new date has been announced, but the game will definitely not be played on the given date. Postponed means that the game has not yet begun, but a new match date on a different day has been set. Counted is used when a game has not yet begun and will not be played in the future, but will be counted due to a sports court ruling. It is also used when a game began but was later stopped and will be counted due to a sports court

	h) no entry	ruling. Counted is also applied when a game was completed in regular fashion but is later counted in accordance with a sports court ruling. Stopped describes when a game is stopped and it has not yet been announced whether it will be counted or whether/when it will be replayed or continued. Replayed is used if a game is replayed that was previously stopped. It is also used if a game is replayed that was previously completed in regular fashion and the date of the replay has been announced. Continued describes when a game was started and will be continued and completed on another day. Annulled is used when a game is completed but will not be counted and it has not yet been announced whether the game will be counted or whether/when it will be replayed.
Official announcement	a) text entry b) no entry	<i>Official announcement</i> gives the official reason for the cancellation, postponement, stoppage, replaying, continuation or annulment of a match.
Time span start date	a) starting date	<i>Time span start date</i> indicates when a match will start if the exact date has not yet been set, but the starting and ending dates of the time span have been announced.
Time span end date	a) ending date	<i>Time span end date</i> indicates when a match will end if the exact date has not yet been set, but the starting and ending dates of the time span have been announced.
Stadium		Information about a <i>stadium</i> .
Stadium ID	a) alphanumeric	<i>Stadium ID</i> is the unique stadium number assigned by the match data collector.
Official name	a) alphanumeric	<i>Official name</i> represents the stadium's <i>Official name</i> .
Street	a) alphanumeric	<i>Street</i> is the street name in the stadium's address.
Number	a) alphanumeric	<i>Number</i> is the number in the stadium's address.
Postal code	a) alphanumeric	<i>Postal code</i> is the postal code in the stadium's address.
Town/city	a) alphanumeric	<i>Town/city</i> is the town or city in the stadium's address.
Country	a) Germany	<i>Country</i> is the country in the stadium's address.

Website	a) alphanumeric	<i>Website</i> is the URL of the stadium's official web page.
Year of construction	a) numeric	<i>Year of construction</i> is the year in which the stadium's construction was completed.
Surface	a) natural grass b) artificial turf	<i>Surface</i> indicates whether the pitch is made of natural grass or artificial turf.
Stadium roof	a) yes b) no	<i>Stadium roof</i> indicates whether the stadium has a fully retractable roof.
Club		Information about a <i>club</i> .
Club ID	a) alphanumeric	<i>Club ID</i> is the unique club number assigned by the match data collector.
Competition	a) reference competition ID	<i>Competition</i> references the competition that the club plays in, with league competitions receiving highest priority.
Stadium	a) reference stadium ID	<i>Stadium</i> references the club's permanent home stadium.
Full legal name incl. legal structure	a) alphanumeric	<i>Full legal name</i> is the club's full legal name, including the club's legal structure.
Unabbreviated form German	a) alphanumeric	<i>Unabbreviated form German</i> is the official unabbreviated form of the club's name in German. It can be found in column "Tabelle (National)" of the DFL's official document "Clubnaming".
Abbreviated form German	a) alphanumeric	<i>Abbreviated form German</i> is the official abbreviated form of the club's name in German. It can be found in column "Clubname Kurz" of the DFL's official document "Clubnaming".
Unabbreviated form	a) alphanumeric	<i>Unabbreviated form</i> is the official unabbreviated form of the club's name in English. It can be found in column "Tabelle (International)" of the DFL's official document "Clubnaming".
Abbreviated form	a) alphanumeric	<i>Abbreviated form</i> is the official abbreviated form of the club's name in English. It can be found in column "Dauerspielstand (International)" of the DFL's official document "Clubnaming".
Official three-letter-code	a) alphanumeric	<i>Official three-letter-code</i> is the club's official three-letter-code. It can be found in column "3-Letter-Code" of the DFL's official document "Clubnaming".

Club colour 1	a) alphanumeric	<i>Club colour 1</i> is the club's first colour.
Club colour 1 - RGB	a) alphanumeric	<i>Club colour 1 – RGB</i> is the club's first colour as an RGB value, in reference to the RGB colour model (R = red, G = green, B = blue; e.g. yellow = 255,255,0)
Club colour 2	a) alphanumeric	<i>Club colour 2</i> is the club's second colour.
Club colour 2 - RGB	a) alphanumeric	<i>Club colour 2 – RGB</i> is the club's second colour as an RGB value, in reference to the RGB colour model (R = red, G = green, B = blue; e.g. yellow = 255,255,0).
Club colour 3	a) alphanumeric	<i>Club colour 3</i> is the club's third colour.
Club colour 3 - RGB	a) alphanumeric	<i>Club colour 3 – RGB</i> is the club's third colour as an RGB value, in reference to the RGB colour model (R = red, G = green, B = blue; e.g. yellow = 255,255,0).
Jersey 1 - main colour	a) alphanumeric	<i>Jersey 1 – main colour</i> is the main colour of a club's official jersey option 1.
Jersey 1 - secondary colour	a) alphanumeric	<i>Jersey 1 – secondary colour</i> is the corresponding secondary colour of a club's official jersey option 1.
Jersey 1 - number colour	a) alphanumeric	<i>Jersey 1 – number colour</i> is the corresponding colour of a club's official jersey option 1, which enables the jersey number to be clearly distinguishable from the jersey's main colour. Usually this refers to the number colour itself.
Jersey 2 - main colour	a) alphanumeric	<i>Jersey 2 – main colour</i> is the main colour of a club's official jersey option 2.
Jersey 2 - secondary colour	a) alphanumeric	<i>Jersey 2 – secondary colour</i> is the corresponding secondary colour of a club's official jersey option 2.
Jersey 2 - number colour	a) alphanumeric	<i>Jersey 2 – number colour</i> is the corresponding colour of a club's official jersey option 2, which enables the jersey number to be clearly distinguishable from the jersey's main colour. Usually this refers to the number colour itself.
Jersey 3 - main colour	a) alphanumeric	<i>Jersey 3 – main colour</i> is the main colour of a club's official jersey option 3.
Jersey 3 - secondary colour	a) alphanumeric	<i>Jersey 3 – secondary colour</i> is the corresponding secondary colour of a club's official jersey option 3.

Jersey 3 - number colour	a) alphanumeric	<i>Jersey 3 – number colour</i> is the corresponding colour of a club's official jersey option 3, which enables the jersey number to be clearly distinguishable from the jersey's main colour. Usually this refers to the number colour itself.
Founding date	a) alphanumeric	<i>Founding date</i> indicates the club's founding date.
Street	a) alphanumeric	<i>Street</i> is the street name in the address of the club's (main) office.
Number	a) alphanumeric	<i>Number</i> is the number in the address of the club's (main) office.
Postal code	a) alphanumeric	<i>Postal code</i> is the postal code in the address of the club's (main) office.
Town/city	a) alphanumeric	<i>Town/city</i> is the town or city in the address of the club's (main) office.
Country	a) Germany	<i>Country</i> is the country in the address of the club's (main) office.
Phone number	a) alphanumeric	<i>Phone number</i> is the club's (main) phone number.
Fax number	a) alphanumeric	<i>Fax number</i> is the club's (main) fax number.
E-mail address	a) alphanumeric	<i>E-mail address</i> is the club's official e-mail address.
Website	a) alphanumeric	<i>Website</i> is the URL of the club's official web page.
Person		Information about a <i>person</i> . The attributes height, weight, jersey number and playing position are to be used for players. In all other cases these attributes are not used.
Person ID	a) alphanumeric	<i>Person ID</i> is the unique person number assigned by the match data collector.
Club	a) reference club ID b) no entry	<i>Club</i> refers to the team that the person is associated with. For referees the manifestation no entry is assigned to this attribute.

Role	a) player b) head coach c) assistant coach d) referee e) team official	<i>Role</i> indicates the role that the person has.
Second role	a) head coach b) assistant coach c) team official d) no entry	<i>Second role</i> indicates the second role that the person has. This attribute is only used if a person has two distinctly different roles at the same time.
Third role	a) head coach b) assistant coach c) team official d) no entry	<i>Third role</i> indicates the third role that the person has. This attribute is only used if a person has three distinctly different roles at the same time.
Full name	a) alphanumeric	<i>Full name</i> is the person's full name, including first name(s) and last name.
First name	a) alphanumeric	<i>First name</i> lists the person's first name(s).
Last name	a) alphanumeric	<i>Last name</i> is the person's last name.
Alias	a) alphanumeric b) no entry	<i>Alias</i> is the person's alias.
Date of birth	a) alphanumeric	<i>Date of birth</i> is the person's exact date of birth, including the day.
Place of birth	a) alphanumeric	<i>Place of birth</i> is the person's place of birth (name of the place at the time of birth).
Geburtsland	a) alphanumeric German	<i>Geburtsland</i> is the country of birth at the time of birth in German.
Country of birth	b) alphanumeric	<i>Country of birth</i> is the country of birth at the time of birth in English.
País de nacimiento	c) alphanumeric Spanish	<i>País de nacimiento</i> is the country of birth at the time of birth in Spanish.

Nationality in accordance with FIFA competitions	a) alphanumeric	<i>Nationality in accordance with FIFA competitions</i> is the citizenship or first citizenship of the country that the person represents or is eligible to play for at FIFA competitions.
Additional nationality 1 (citizenship)	a) alphanumeric b) no entry	<i>Additional nationality 1</i> is the person's additional citizenship.
Additional nationality 2 (citizenship)	a) alphanumeric b) no entry	<i>Additional nationality 2</i> is the person's additional citizenship.
Height	a) numeric b) no entry	<i>Height</i> is the height of a player in centimetres.
Weight	a) numeric b) no entry	<i>Weight</i> is the weight of a player in kilograms.
Homegrown player	a) club product b) association product c) no entry	Indicates whether a player is a club or association product according to §5 b paragraphs 1 and 2 of the player licensing regulations. If the player is not classified as homegrown the manifestation no entry is used.
Suspension		Information about a player, who has been officially suspended from taking part in at least one game organised by the DFL due to a rule violation, disciplinary reasons or other irregularities. The applicable statutes can be found in the responsible leagues' and associations' respective legal and procedural regulations.
Type of suspension	a) number of matches DFL competitions b) number of matches DFB sector c) time period	Indicates whether a player is suspended for a certain number of games in a particular competition, or for a certain period of time.
Number of matches	a) numeric b) no entry	Indicates the <i>number of matches</i> a player is suspended for. If a player's suspension is set for a period of time and not for a number of matches, the manifestation no entry is used.
Entity awarding suspension	a) DFB/DFL b) FIFA/International associations c) other	Refers to the entity awarding the suspension.
Scope of suspension	a) DFL competition b) DFB sector c) DFB and international games	Indicates the <i>scope of a player's suspension</i> . DFL competition indicates that the player is only suspended from games organised by the DFL. DFB sector indicates that the player is suspended from games and competitions at national level within the DFB sector (e.g. DFL games and DFB cup games). DFB and international games means that besides being

	d) worldwide e) other	suspended from national competitions, the player is also suspended from playing international games for teams within the DFB sector (e.g. international games, European cup games of DFB teams).
Start of suspension	a) date	Indicates the <i>start of a player's suspension</i> . This is usually the day on which the rule violation that led to the suspension is first penalised or the day on which the suspension is first issued. → note: a case can go through several instances of appeal and therefore there can be several days on which judgements are passed.
End of suspension	a) date b) no entry	Indicates the <i>end of a player's suspension</i> . If a player is suspended for a certain number of games the suspension usually ends on the day of the last game the suspension applies to. If the end of the suspension has not (yet) been announced or if the suspension is lifted, the manifestation no entry is used.
Reason for suspension	a) 5 th yellow card b) 10 th yellow card c) 15 th yellow card d) 20 th yellow card e) 25 th yellow card f) second yellow card in the same match g) straight red card (DFL competition) h) straight red card (outside of DFL) i) other offence during the game (DFL competition) j) other offence during the game (outside of DFL) k) playing without eligibility l) no clearance eligibility m) betting, manipulation, bribery n) discrimination o) doping p) other	Gives the <i>reason for the suspension</i> . If a suspension takes effect based on a sports court ruling, the reason that is either given in the ruling itself or which comes closest to it must be entered here.
Lifting of suspension	a) date b) no entry	Indicates the point in time when an issued suspension is lifted with immediate effect before the intended end of the suspension.
Injury		Describes a situation in which a player is unlikely to be able to take part in matches or his participation appears to be at risk due to his state of health or fitness.
Start of injury	a) date	Indicates the point in time after which a player's ongoing state of injury is first announced and recorded.

End of injury	a) date b) no entry	Indicates the point in time after which a player's ongoing state of injury is considered to have ended. The manifestation no entry is used until the end of the injury has been determined and recorded.
Art Verletzung	a) text entry	<i>Art Verletzung</i> indicates the type of injury or illness, or the state of the player (e.g. training deficit, rehab training) in German. If the type of injury changes during the ongoing state of injury, the type is updated accordingly, but no new injury is recorded.
Kind of injury	a) text entry	<i>Kind of injury</i> indicates the type of injury or illness, or the state of the player in English.
Injury status	a) injured b) doubtful	Indicates whether a player is not expected to feature in a game at all or whether his participation is merely in doubt.
Jersey number	a) numeric b) no entry	<i>Jersey number</i> refers to the jersey number a player wears.
Spielposition	a) Torwart b) Verteidigung c) Mittelfeld d) Angriff e) no entry	<i>Spielposition</i> indicates the basic playing position in German that a player is assigned by his club at the beginning of a season. Should this position change permanently for a longer period of time during a season, the entry is altered.
Playing position	a) goalkeeper b) defense c) midfield d) offense e) empty	<i>Playing position</i> indicates the basic playing position in English that a player is assigned by his club at the beginning of a season. Should this position change permanently for a longer period of time during a season, the entry is altered.
Posicion	a) Portero b) Defensa c) Medio Campo d) Ataque e) vacio	<i>Posicion</i> indicates the basic playing position in Spanish that a player is assigned by his club at the beginning of a season. Should this position change permanently for a longer period of time during a season, the entry is altered.

3. MATCH DATA

3.1 MATCH INFORMATION

General		<i>General information about a game.</i>
Identification number	a) reference identification number	<i>Identification number</i> is the unique match number assigned by the match data collector.
Official match number	a) reference official match number	<i>Official match number</i> is the match number assigned by the organiser in reference to the stage of the season.
Actual kick-off time	a) time stamp	Returns the actual date and time when the match started in the UTC format.
Environment		Information about the <i>environment</i> of a game.
Stadium	a) reference stadium ID	<i>Stadium</i> is the stadium in which the match will be played.
Stadium roof	a) open b) closed	<i>Stadium roof</i> indicates whether the (retractable) stadium roof is open or closed approx. 30 minutes before the start of the match. If the status of the roof changes after the time of recording, the status that prevailed for the majority of the match must be recorded.
Flood lights	a) switched on as necessary b) switched on c) switched off	<i>Flood lights</i> indicates whether the flood lights are switch on or off approx. 30 minutes before the start of the match. If the flood lights are switched on, a distinction is made whether the environmental conditions render it necessary to play the game under flood lights or whether they have been switched on for other reasons.
Pitch length	a) numeric	<i>Pitch length</i> is the length of the pitch in metres, correct to one decimal place.
Pitch width	a) numeric	<i>Pitch width</i> is the width of the pitch in metres, correct to one decimal place.

Pitch condition	a) no pitch wear b) slight pitch wear c) medium pitch wear d) severe pitch wear	<i>Pitch condition</i> is the level of pitch surface deterioration determined by the match data collector. The surface either exhibits no worn patches (no pitch wear), small worn patches (slight pitch wear), larger worn patches (medium pitch wear) or large areas of worn patches (severe pitch wear).
Stadium capacity	a) numeric	<i>Stadium capacity</i> is the number of spectator seats permitted in the stadium.
Attendance	a) numeric	<i>Attendance</i> is the number of spectators inside the stadium according to the display/announcement.
Use to capacity	a) sold out b) not sold out	<i>Use to capacity</i> indicates whether the stadium has been sold out or not according to the display/announcement.
Precipitation	a) no precipitation b) light rain c) heavy rain d) light snowfall e) heavy snowfall	<i>Precipitation</i> indicates whether – and if so, how heavily – it is raining or snowing approx. 15 minutes before the start of the game. If the weather changes significantly after the time of recording, the degree of precipitation that prevailed for the majority of the match, or the degree of precipitation that influenced the match most heavily according to the match data collector, must be recorded.
Temperature	a) numeric	<i>Temperature</i> indicates the temperature in °C inside the stadium approx. 15 minutes before the start of the game. It is to be obtained from a mobile weather station at the recording location by the match data collector.
Humidity	a) numeric	<i>Humidity</i> indicates the humidity in % inside the stadium approx. 15 minutes before the start of the game. It is to be obtained from a mobile weather station at the recording location by the match data collector.
Atmospheric pressure	a) numeric	<i>Atmospheric pressure</i> is the atmospheric pressure in hPa inside the stadium approx. 15 minutes before the start of the game. It is to be obtained from a mobile weather station at the recording location by the match data collector.
Teams		Information about the participating teams. The attributes player, jersey number, playing position, team captain and starting line-up of every player in the starting line-up and of the substitutes are to be recorded.
Home team	a) reference club ID	<i>Home team</i> references the home team assigned by the organiser.
Away team	a) reference club ID	<i>Away team</i> references the away team assigned by the organiser.
Players	a) reference person ID	<i>Players</i> lists the players in the starting line-up and the substitutes that were reported to the referee before the start of the game in accordance with the DFB football rules.

Jersey number	a) numeric	<i>Jersey number</i> indicates the jersey numbers assigned to the players in the starting line-up and to the substitutes that were reported to the referee before the start of the game in accordance with the DFB football rules. In the exceptional case that a player lines up with a jersey number that is not assigned to him, the jersey number that he is wearing is to be recorded.
Starting line-up	a) yes b) no	<i>Starting line-up</i> indicates whether the player in question is part of the starting line-up at kick-off.
Playing position (only used for substitutes in the case of them being brought on)	a) GK: goalkeeper b) LB: left back c) LCB: left centre back d) CB: centre back e) RCB: right centre back f) RB: right back g) WLDM: wide-left defensive midfielder h) LDM: left defensive midfielder i) CDM: central defensive midfielder j) RDM: right defensive midfielder k) WRDM: wide-right defensive midfielder l) LM: left midfielder m) HLM: half-left midfielder n) CM: central midfielder o) HRM: half-right midfielder p) RM: right midfielder q) LAM: left attacking midfielder r) HLAM: half-left attacking midfielder s) CAM: central attacking midfielder t) HRAM: half-right attacking midfielder u) RAM: right attacking midfielder v) SS: supporting striker w) LW: left wide player x) LS: left striker y) CF: centre forward	<i>Playing position</i> is the playing position assigned to the players in the starting line-up and to the substitutes. The match data collector is responsible for correctly recording the playing positions. The playing position of substitutes is only to be recorded for players that are actually brought on during the course of the match and whose playing position can be conclusively determined (for example, this is possibly difficult for players that are only brought on for a few minutes; in this case the playing position of the player that was taken off is to be assumed as correct and recorded).

z) RS: right striker
aa) RW: right wide player

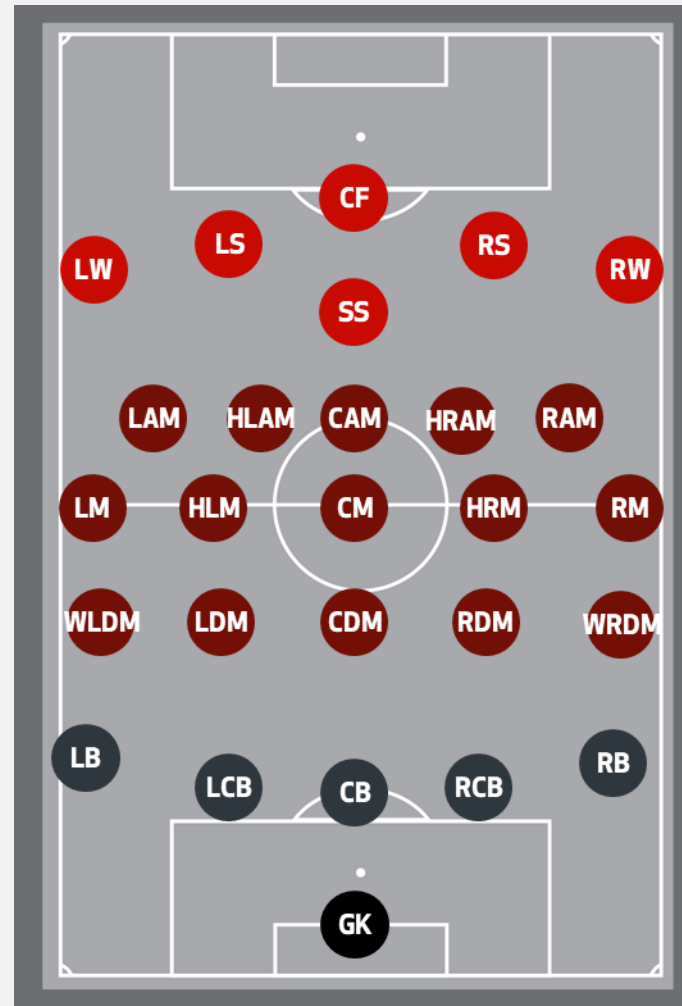


Figure 3: Playing positions and their abbreviations

Team captain	a) yes b) no	<i>Team captain</i> is the captain of a team that has been reported to the referee before the start of the game in accordance with the DFB football rules, or the player wearing the captain's armband at the start of the game.
Head coach	a) reference person ID	<i>Head coach</i> is the respective head coach of the home or away team.
Assistant coach	a) reference person ID	<i>Assistant coach</i> is the respective assistant coach of the home or away team.
Acting coach	a) reference person ID	<i>Acting coach</i> indicates the respective member of the coaching staff of the home or away team standing in for the head coach at the side of the pitch. Whenever the head coach takes charge of his team at the side of the pitch the attribute acting coach is left blank. A change of acting coach after kick-off due to the original acting coach being sent off is not to be taken into account.
Team official person	a) reference person ID	<i>Team official person</i> refers to a team official of the home or away team.
Team official function	a) doctor b) massage therapist c) kit manager d) team manager e) first official f) second official g) additional official	<i>Team official function</i> indicates the function the respective team official of the home or away team fulfills.
Tactical formation at the start of the game	a) 4-2-3-1 b) 4-1-4-1 c) 4-1-4-1 with an attacking midfield row d) 4-1-4-1 normal wide players, two central attacking midfielders e) 4-4-2 with a double pivot f) 4-4-2 with a double pivot and attacking wide players g) 4-4-2 with a diamond h) 4-3-3 i) 4-3-3 with one defensive midfielder j) 4-3-3 with a double pivot k) 4-3-1-2 l) 4-5-1	<i>Tactical formation</i> indicates which tactical formation the home or away team have chosen at the start of the game. The manifestations are described in detail below: 4-2-3-1: GK – LB, LCB, RCB, RB – LDM, RDM – LAM, CAM, RM – CF 4-1-4-1: GK – LB, LCB, RCB, RB – CDM – LM, HLM, HRM, RM – CF 4-1-4-1 with an attacking midfield row: GK – LB, LCB, RCB, RB – CDM – LAM, HLAM, HRAM, RM – CF 4-1-4-1 normal wide players, two central attacking midfielders: GK – LB, LCB, RCB, RB – CDM – LM, HLM, HRAM, RM – CF 4-4-2 with a double pivot: GK – LB, LCB, RCB, RB – LM, LDM, RDM, RM – LS, RS 4-4-2 with a double pivot and attacking wide players: GK – LB, LCB, RCB, RB – LAM, LDM, RDM, RM – LS, RS 4-4-2 with a diamond: GK – LB, LCB, RCB, RB – HLM, CDM, CAM, HRM – LS, RS 4-3-3: GK – LB, LCB, RCB, RB – HLM, CM, HRM – LW, CF, RW 4-3-3 with one defensive midfielder: GK – LB, LCB, RCB, RB – HLM, CDM, HRM – LW, CF, RW 4-3-3 with a double pivot: GK – LB, LCB, RCB, RB – LDM, CAM, RDM – LW, CF, RW 4-3-1-2: GK – LB, LCB, RCB, RB – LDM, CDM, RDM – CAM – LS, RS 4-5-1: GK – LB, LCB, RCB, RB – LM, LDM, CM, RDM, RM – CF

m) 4-5-1 with attacking wide players n) 5-3-2 o) 5-3-2 with one central defensive midfielder p) 5-3-1-1 q) 5-3-1-1 with one central defensive midfielder r) 5-4-1 s) 5-4-1 with attacking wide players t) 5-4-1 with a diamond u) 3-5-2 v) 3-4-3 w) 3-4-3 with a defensive midfield row x) 3-4-3 with a diamond y) 3-3-2-2 z) 3-3-2-2 with two central attacking midfielders aa) 3-5-1-1 ab) 3-5-1-1 with defending wide players ac) 3-4-2-1 ad) 3-4-2-1 with a defensive midfield row ae) 3-4-2-1 with two central attacking midfielders af) 3-4-1-2 ag) 3-4-1-2 with a defensive midfield row ah) 4-4-1-1 ai) 4-4-1-1 with attacking wide players aj) 4-3-2-1 ak) 4-1-3-2 al) 3-1-4-2 am) 3-5-2 with attacking wide players an) 3-3-3-1 ao) 5-3-2 with a double pivot	4-5-1 with attacking wide players: GK – LB, LCB, RCB, RB – LAM, HLM, CDM, HRM, RAM – CF 5-3-2: GK – LB, LCB, CB, RCB, RB – LM, CM, RM – LS, RS 5-3-2 with one central defensive midfielder: GK – LB, LCB, CB, RCB, RB – HLM, CDM, HRM – LS, RS 5-3-1-1: GK – LB, LCB, CB, RCB, RB – LM, CM, RM – SS - CF 5-3-1-1 with one central defensive midfielder: GK – LB, LCB, CB, RCB, RB – LM, CDM, RM – SS - CF 5-4-1: GK – LB, LCB, CB, RCB, RB – LM, LDM, RDM, RM – CF 5-4-1 with attacking wide players: GK – LB, LCB, CB, RCB, RB – LAM, LDM, RDM, RAM – CF 5-4-1 with a diamond: GK – LB, LCB, CB, RCB, RB – HLM, CDM, CAM, HRM – CF 3-5-2: GK – LCB, CB, RCB – LM, LDM, CM, RDM, RM – LS, RS 3-4-3: GK – LCB, CB, RCB – LM, LDM, RDM, RM – LW, CF, RW 3-4-3 with a defensive midfield row: GK – LCB, CB, RCB – WLDM, LDM, RDM, WRDM – LW, CF, RW 3-4-3 with a diamond: GK – LCB, CB, RCB – LM, CDM, CAM, RM – LW, CF, RW 3-3-2-2: GK – LCB, CB, RCB – WLDM, CDM, WRDM – HLM, HRM – LS, RS 3-3-2-2 with two central attacking midfielders: GK – LCB, CB, RCB – LM, CDM, RM – HLAM, HRAM – LS, RS 3-5-1-1: GK – LCB, CB, RCB – LM, LDM, CM, RDM, RM – SS - CF 3-5-1-1: with defending wide players: GK – LCB, CB, RCB – WLDM, LDM, CM, RDM, WRDM – SS - CF 3-4-2-1: GK – LCB, CB, RCB – LM, LDM, RDM, RM – LAM, RAM – CF 3-4-2-1 with a defensive midfield row: GK – LCB, CB, RCB – WLDM, LDM, RDM, WRDM – LAM, RAM – CF 3-4-2-1 with two central attacking midfielders: GK – LCB, CB, RCB – WLDM, LDM, RDM, WRDM – HLAM, HRAM – CF 3-4-1-2: GK – LCB, CB, RCB – LM, LDM, RDM, RM – CAM – LS, RS 3-4-1-2 with a defensive midfield row: GK – LCB, CB, RCB – WLDM, LDM, RDM, WRDM – CAM – LS, RS 4-4-1-1: GK – LB, LCB, RCB, RB – LM, LDM, RDM, RM – SS - CF 4-4-1-1 with attacking wide players: GK – LB, LCB, RCB, RB – LAM, LDM, RDM, RAM – SS - CF 4-3-2-1: GK – LB, LCB, RCB, RB – LM, CM, RM – LAM, RAM – CF 4-1-3-2: GK – LB, LCB, RCB, RB – CDM - LAM – CAM – RAM - LS, RS 3-1-4-2: GK – LCB, CB, RCB – CDM - LM – HLM – HRM – RM - LS, RS 3-5-2 with attacking wide players: GK – LCB, CB, RCB - LAM – HLM – CDM – HRM – RAM - LS, RS 3-3-3-1: GK – LCB, CB, RCB – LDM – CDM – RDM - LAM – CAM – RAM – CF 5-3-2 with a double pivot: GK – LB, LCB, CB, RCB, RB – LDM – RDM – CAM - LS, RS The match data collector must use one of the specified formations. If none of the described formations apply, the data collector is to select the specified formation that comes closest to the formation being played by the team in question.
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Jersey option: number - home	a) alphanumeric b) no entry	<i>Jersey option: number - home</i> refers to the home team's jersey option stored in the master data. If no jersey option number has been stored in the master data, the manifestation no entry is used.
Jersey option: number - away	a) alphanumeric b) no entry	<i>Jersey option: number - away</i> refers to the away team's jersey option stored in the master data. If no jersey option number has been stored in the master data, the manifestation no entry is used.
Jersey home - main colour	a) alphanumeric b) no entry	<i>Jersey home - main colour</i> refers to the home team's main jersey colour worn by the outfield players at the start of the game.
Jersey home - secondary colour	a) alphanumeric b) no entry	<i>Jersey home - secondary colour</i> refers to the home team's secondary jersey colour worn by the outfield players at the start of the game.
Jersey home - number colour	a) alphanumeric b) no entry	<i>Jersey home - number colour</i> refers to the colour worn by the home team's outfield players at the start of the game, which enables the jersey number to be clearly distinguishable from the jersey's main colour. Usually this refers to the number colour itself.
Jersey away - main colour	a) alphanumeric b) no entry	<i>Jersey away - main colour</i> refers to the away team's main jersey colour worn by the outfield players at the start of the game.
Jersey away - secondary colour	a) alphanumeric b) no entry	<i>Jersey away - secondary colour</i> refers to the away team's secondary jersey colour worn by the outfield players at the start of the game.
Jersey away - number colour	a) alphanumeric b) no entry	<i>Jersey away - number colour</i> refers to the colour worn by the away team's outfield players at the start of the game, which enables the jersey number to be clearly distinguishable from the jersey's main colour. Usually this refers to the number colour itself.
Referees		Information about the participating <i>referees</i> .
Referee	a) reference person ID	<i>Referee</i> indicates the referee in charge of the match at the start of the game.
Assistant Referee 1	a) reference person ID	<i>Assistant referee 1</i> refers to the assistant referee that has taken up a position on the touchline closest to the substitutes' bench at the start of the game.
Assistant Referee 2	a) reference person ID	<i>Assistant referee 2</i> refers to the assistant referee that has taken up a position on the touchline opposite the substitutes' bench at the start of the game.

Video assistant referee	a) reference person ID	<i>Video assistant referee</i> refers to the game's video assistant referee.
Fourth official	a) reference person ID	<i>Fourth official</i> refers to the game's fourth official at the start of the game.
Other		Additional, noteworthy information about the match.
Other information	a) text entry	<i>Other information</i> gives additional, noteworthy match information, which either leads to a significant delay of the kick-off or which has other far-reaching effects on the proceedings. A delay is deemed to be significant if kick-off is delayed by more than 5 minutes. Noteworthy and far-reaching events are, for example, a goal falling over, flood lights failing before the start of the game, a team or a referee showing up late or a delay in kick-off due to severe traffic, weather conditions or security concerns (e.g. spectators outside the stadium, faulty stadium roof, etc.). In addition, reasons for the abandonment of a game or for playing a game behind closed doors, for example, can be listed here.

3.2 MATCH EVENTS

Ball played		<i>A ball played</i> describes a player's attempt to switch ball control to another member of his team.
Team	a) reference club ID	<i>Team</i> refers to the team that attempted to play the ball.
Player	a) reference person ID	<i>Player</i> refers to the person who attempted to switch ball control to another member of his team.
Origin of ball played	a) own half b) opponents' half c) no entry	<i>Origin of ball played</i> refers to the location on the pitch from where the ball was played.
Recipient	a) reference person ID b) no entry	<i>Recipient</i> refers to the person who is assigned the subsequent ball action following the ball played. If the game is halted after the ball was played, no recipient is recorded.
First phase of play (goalkeeper)	a) kick-out b) throw-out c) no entry	<i>First phase of play (goalkeeper)</i> indicates whether the goalkeeper executed a throw-out or a kick-out (from his hands).

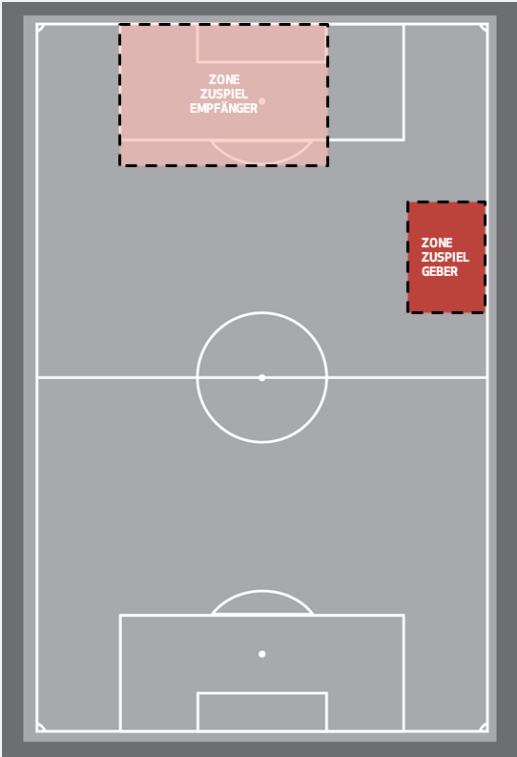
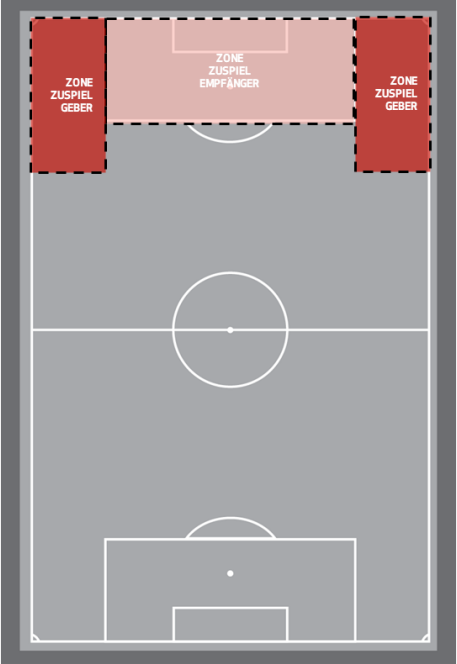
Execution type	a) pass b) cross	<i>Execution</i> type indicates whether the ball was played as a pass or as a cross.
Half-field ball into the area	a) yes b) no	<i>Half-field ball into the area</i> is a specification of the ball action ball played, for which the positions of the player playing the ball into the area and of the possible recipient are determined. The player playing the ball must be located approximately inside one of the zones marked by the extended side of the penalty area, an imaginary horizontal line at the end of the centre circle, the touchline and an imaginary quarter-way line, which would be drawn at a quarter of the length of the pitch parallel to the half-way line (see figure 3a). A possible recipient must be located approximately inside the zone from the far side of the penalty area to the “near” post and from the goal line to furthest point of the penalty arc. Furthermore, the distance of the ball played must be medium-length or long (attribute <i>distance</i>) and the height of the ball played must be high (attribute <i>height</i>). In order to classify a ball played as a half-field ball into the area, if the ball is blocked by an opposing player, it is not the actual distance travelled that is decisive, but the intended distance. 

Figure 3a: Overview of zones for half-field balls into the area		
Low ball into the area	a) yes b) no	<i>Low ball into the area</i> is a specification of the ball action ball played, for which the positions of the player playing the ball into the area and of the possible recipient are determined. The player playing the ball must be located approximately inside one of the zones marked by the extended sides of the penalty area, the goal line, the touchlines and an imaginary quarter-way line, which would be drawn at a quarter of the length of the pitch parallel to the half-way line (see figure 3b). The possible recipient must be located approx. inside the penalty area. Furthermore, the distance of the ball played must be medium-length or long (attribute <i>distance</i>) and the height of the ball played must be low (attribute <i>height</i>). In order to classify a ball played as a low ball into the area if the ball is blocked by an opposing player, it is not the actual distance travelled that is decisive, but the intended distance. A low ball into the area played from a <i>baseline crossing zone</i> is called a <i>low baseline ball into the area</i>.  Figure 3b: Overview of zones for low balls into the area
From open play	a) yes b) no	<i>From open play</i> indicates whether the ball was played from open play. If this is not the case, the manifestation no is used.

Phase of possession number	a) numeric	<i>Phase of possession number</i> specifies the phase of possession in question.
Evaluation	a) successfully received b) successful c) not successful	<i>Evaluation</i> indicates the success of a ball played. A ball played classified as a pass in the execution type attribute and as not successful in the evaluation attribute, is called a misplaced pass. If the team of the player receiving the ball and that of the player playing the ball are identical, the ball played is classified as successfully received. If the player receiving the ball is on the opposing team of the player playing the ball, the ball played is classified as not successful. If the team playing the ball stays in possession despite no player having received the ball, the ball played is classified as successful. This is the case, for example, if a player of the team playing the ball is fouled before he can successfully receive the ball. If no player receives the ball and the team that played the ball does not stay in possession, the ball played is classified as not successful.
Height	a) low b) high	<i>Height</i> indicates whether the ball is played above or below knee height. For the classification of the height of blocked crosses, it is not the actual height of the ball that is decisive, but the intended height.
Distance	a) short b) medium length c) long	<i>Distance</i> indicates whether the ball travels less than 10 metres (short), 10 to 30 metres (medium length) or more than 30 metres (long). For crosses only the manifestations medium length or long are to be used. For the classification of distance, it is not the actual distance travelled that is decisive, but the intended distance.
Rotation	a) inswinging b) outswinging c) other d) no entry	<i>Rotation</i> indicates whether the ball's trajectory is being significantly altered by its lateral rotation - either towards goal (inswinging) or away from goal (outswinging) - or whether the ball is played in any other way. If a pass has been recorded for the attribute <i>execution type</i> and the manifestation no has been recorded for the attribute <i>ball into the area from half-field</i> , the manifestation no entry is used.
Penalty area	a) yes b) no	The manifestation yes is used for the attribute <i>penalty area</i> if the ball is played into or from within the opponents' penalty area. In all other cases the manifestation no is used.
One-two	a) initiating player b) returning player c) both d) no entry	A <i>one-two</i> is an interaction between two players of the same team involving two consecutive passes during which the ball is first played from player A to player B and from player B straight back to player A. Player B may only touch the ball once. However, such an interaction is only classified as a one-two if at least one player of the opposing team, who is positioned within the triangle created by the pass and the run, is rounded during the interaction. Rounded means that player A must get in behind an opposing player with the ball through the one-two. For one-tuos it is player A's personal point of view that is decisive. This means the rounding can also occur in a sideways or backward direction. If a one-two is deemed to have been played, the first pass from player A to player B is to be recorded as "initiating player", whilst the one-touch return ball from player B to player A is to be recorded as "returning player".
Free kick layoff	a) yes b) no	If the sole tactical plan of action at a free kick is to play the ball to a teammate (once or several times) over a short distance to set that teammate up for a shot, then those ball actions are recorded as passes with the specification <i>free kick layoff</i> . In these cases, stopping the ball is recorded as a ball played and as a pass in the execution type attribute. A <i>free kick layoff</i> is not recorded if the tactical plan of action is to significantly change the match situation at the free kick (e.g. by playing the ball past the wall).

Passing angle	a) alphanumeric b) no entry	<i>Passing angle</i> indicates the angle of the pass in ° (degrees). The angle is calculated from the X and Y positions of the pass as well as the X and Y source positions of the subsequent match action. 0° serves as a reference line and is defined as the imaginary line from the kick-off spot to the penalty spot in the direction of play. The number of degrees increases in a clockwise direction. If no position is recorded for one or more subsequent match actions, the next action with a recorded position is to be used.
Direction	a) forward (0°-22° & 338°-360°) b) diagonal (23°-67° & 293°-337°) c) sideways (68°-112° & 248°-292°) d) backward (113° & 247°) e) no entry	<i>Direction</i> indicates the direction of a long pass from open play, taking the passing angle into consideration. This attribute is only used for short or medium length passes if the pass represents the situation before a shot, an own goal or a chance without a resulting shot.

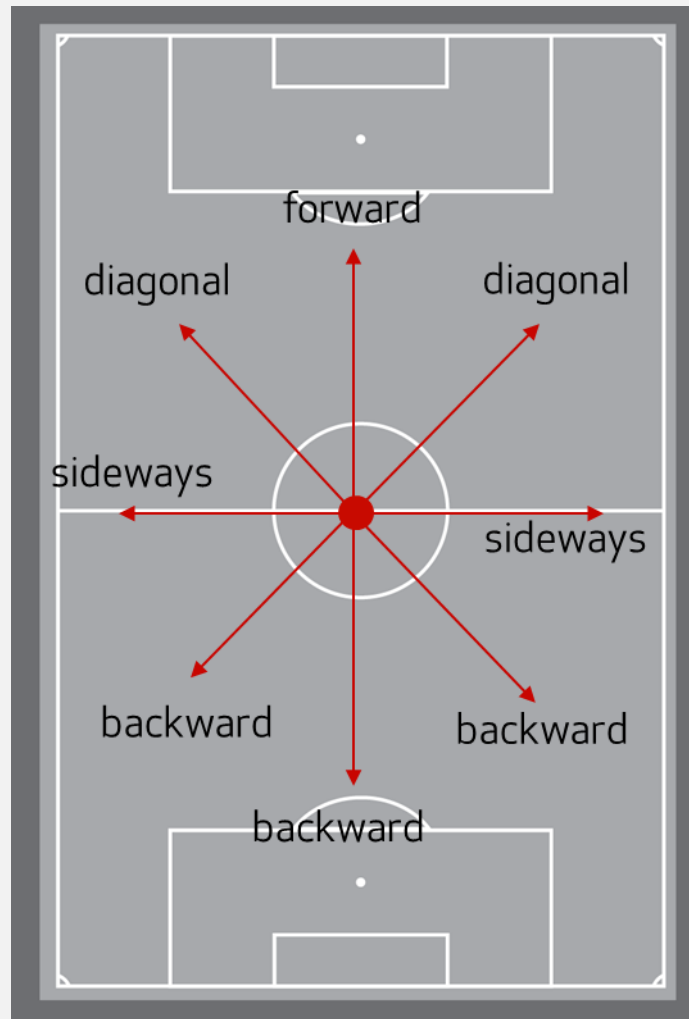


Figure 4: Overview of possible directions for long passes from open play

Cross

Cross is a specification of the ball action ball played, for which the positions of the player playing the cross and of the possible cross recipient are determined. The player playing the cross must be located approx. inside one of the four crossing zones. The four zones are marked by the touchlines, the extended sides of the penalty area, the goal lines and the imaginary quarter-way lines, which would be drawn at a quarter of the length of the pitch parallel to the half-way line (see figure 5). The possible cross recipient must be located approx. inside the penalty area. Furthermore, the distance of the ball played must be medium-length or long (attribute distance) and the height of the ball played must be high (attribute height). In order to classify a ball played as a cross if the ball is blocked by an opposing player, it is not the actual distance travelled that is decisive, but the intended distance.

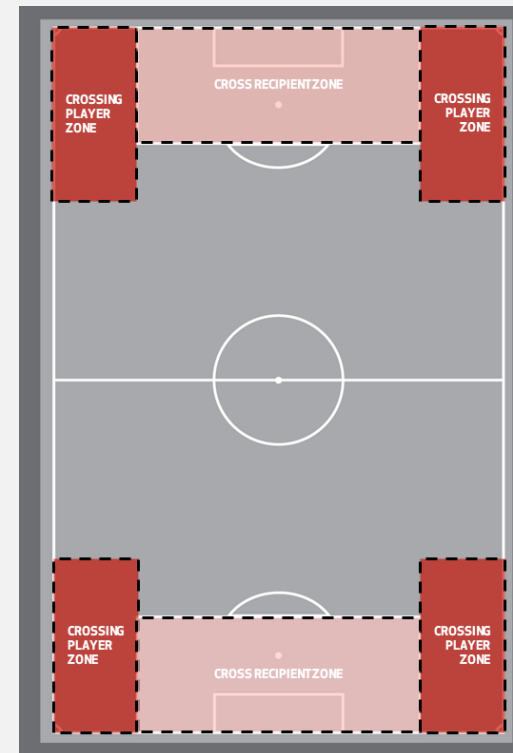
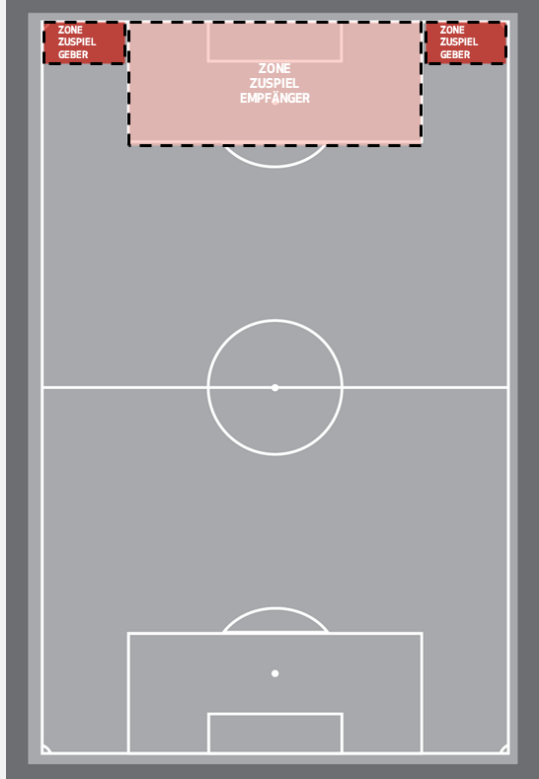


Figure 5: Overview of crossing zones

Baseline cross	a) yes b) no	<i>Baseline cross</i> is a specification of the ball action ball played, for which the positions of the player playing the cross and of the possible recipient are determined. The player playing the baseline cross must be located approximately inside one of the zones marked by the extended sides of the penalty area, the goal line, the touchline and an imaginary line corresponding to the opponents' goal area (see figure 5a). The possible recipient must be located approx. inside the penalty area. Furthermore, the distance of the ball played must be medium-length or long (attribute <i>distance</i>) and the height of the ball played must be high (attribute <i>height</i>). In order to classify a ball played as a baseline cross, if the ball is blocked by an opposing player, it is not the actual distance travelled that is decisive, but the intended distance.  Figure 5a: Overview of baseline crossing zones
Execution side	a) left b) right	<i>Execution side</i> indicates the side of the pitch from which the ball was crossed, from the point of view of the executing team's direction of play.

Goalkeeper action	a) held b) punched c) other goalkeeper touch d) no entry	<i>Goalkeeper action</i> indicates whether the goalkeeper of the team without ball control catches, punches or deflects the ball. The attribute goalkeeper action is only to be used if the ball action succeeding a cross is performed by the goalkeeper of the team without ball control. If the attribute is used, an additional match event ball played, other ball action or ball won is awarded to the goalkeeper, depending on the action that took place.
Goalkeeper	a) reference person ID b) no entry	<i>Goalkeeper</i> refers to the goalkeeper of the team without ball control, who catches, punches or deflects the ball. The attribute is only to be used if the ball action succeeding a cross is performed by the goalkeeper of the team without ball control.
Shot		<i>Shot</i> represents a ball action during which a player in ball control attempts to score a goal. A shot is also recorded if the game is halted due to a rule violation following the shot, for example because a teammate of the player taking the shot was offside. However, no shot is recorded if the player taking the shot commits a rule violation, for example if the shot itself is ruled to have been handball. If a ball played ends up in the goal or hits the goal frame, it is assumed that the player that played the ball had the secondary intention of taking a shot. In this case a shot is recorded, not a ball played.
Team	a) reference club ID	<i>Team</i> refers to the team that took the shot.
Player	a) reference person ID	<i>Player</i> refers to the player that took the shot.
Execution type	a) successful shot b) shot saved c) shot blocked d) shot off the goal frame e) shot wide f) other shot	<i>Execution type</i> indicates whether a shot was successful, saved or blocked, came back off the goal frame (one of the two posts or the crossbar), went wide or took another course.
Shot setup player	a) reference person ID b) no entry	<i>Shot setup player</i> refers to the person that set up the shot, meaning the last teammate of the shot taker to touch the ball before the shot. Exceptions include a touch of the ball - after which the opposing team gains certain ball control before the shot taker receives it; if a player clears the ball, he is only deemed to have had certain ball control if he had the option of playing the ball to a teammate instead of clearing it. - after which the game is halted by the referee before the shot taker receives the ball (e.g. in cases where penalty kicks, free kicks or corner kicks are directly executed as a shot).
Contribution to a shot	a) reference person ID b) no entry	<i>Contribution to a shot</i> is awarded to a player of the team that took the shot, if he is not the shot setup provider, but has made a significant contribution to setting up the shot through his action. This is only the case if his action was performed with the primary intention of contributing to a shot and is applicable to all shot execution types.

		In particular, a contribution to a shot is awarded for the following actions: - the execution of a set piece clearly contributed to the creation of the shot through its type and quality; - the player let the ball through for the shot taker without touching the ball himself, thereby significantly contributing to the creation of the shot; - the player's ball action directly enabled the shot setup provider to set up the shot. A contribution to a shot is only awarded for the execution of a set piece if the type and quality of the execution clearly contributed to the execution of the shot. Per shot, only one person may be awarded a contribution to the shot. A player could also be awarded a contribution to a shot that he took himself, but only if a teammate's (shot setup provider's) action occurred between his contributing action and the taking of the shot.
Shot setup type	a) direct b) indirect c) other d) free kick e) no entry	<i>Shot setup type</i> refers to the type of shot setup. A distinction is made between the following cases: A direct shot setup is a ball action executed within a short period of time (usually a few seconds) before the shot, with the shot taker being the recipient of the ball. Touches of the ball by opposing players after the setup has been executed are considered to be insignificant, provided the shot taker would presumably have gained ball control in a comparable match situation without these touches occurring. An indirect shot setup is a ball action during which the direction of the ball is significantly affected by an opposing player before reaching the shot taker, provided the shot taker would presumably not have gained ball control in a comparable match situation without these touches occurring. An indirect shot setup is also recorded if the shot hits the goal frame, is blocked by an opposing player or is saved by the goalkeeper before reaching the shot taker. If, in this case, the shot taker and the player that took the first shot are one and the same, the indirect shot setup is awarded to the player who performed the last ball action before the first shot (usually the player that played the final ball before the first shot). Other indicates that the ball was played directly to the shot taker by a teammate, whose primary intention was not to set up a shot (e.g. a pass to the shot taker in the team's own half, after which the shot taker goes on an individual run with the ball before scoring; a player is hit by a shot without becoming active; a teammate's shot directly reaches the shot taker). A free kick shot setup indicates that the shot taker receives the ball from a free kick layoff at a free kick. If the shot is a follow-up shot after a penalty kick or a free kick executed as a shot, the following rules apply: If the shot taker is the player that took the penalty kick or free kick, a fouled player shot setup is also awarded for the follow-up shot, provided the ball was not touched by any of the shot taker's teammates before the follow-up shot (e.g. if a teammate in the wall is hit by the initial shot). If the follow-up shot taker is not the player that took the penalty kick

		or free kick, then the player that took the initial shot from the set piece is awarded an indirect shot setup, provided the ball was not touched by any of the shot taker's teammates before the follow-up shot.
Fouled player shot setup	a) reference person ID b) no entry	<i>Fouled player shot setup</i> indicates, when a shot results from a penalty kick or free kick, which was awarded due to a foul against an opposing player. This applies to free kicks that are directly executed as shots or when a free kick layoff leads to a shot. In these cases, the attribute fouled player shot setup refers to the player that was fouled to win the set piece in question. If the manifestation no entry has been used for the attribute shot setup player, then the fouled player can be deemed to have set up the shot (depending on the understanding of what constitutes a setup). This is also the case if the manifestation free kick has been used for the attribute shot setup type.
Phase of possession number	a) numeric	<i>Phase of possession number</i> returns the phase of possession in question.
Shot type	a) left leg b) right leg c) head d) torso	<i>Shot type</i> indicates whether the shot was taken with the left leg or right leg, or through the use of the head or torso. If a shot occurs through the use of the arm or hand, then the shot type torso is to be used.
Shot origin	a) numeric	<i>Shot origin</i> refers to the location on the pitch from where the shot was taken from the point of view of the team in ball control.

Working

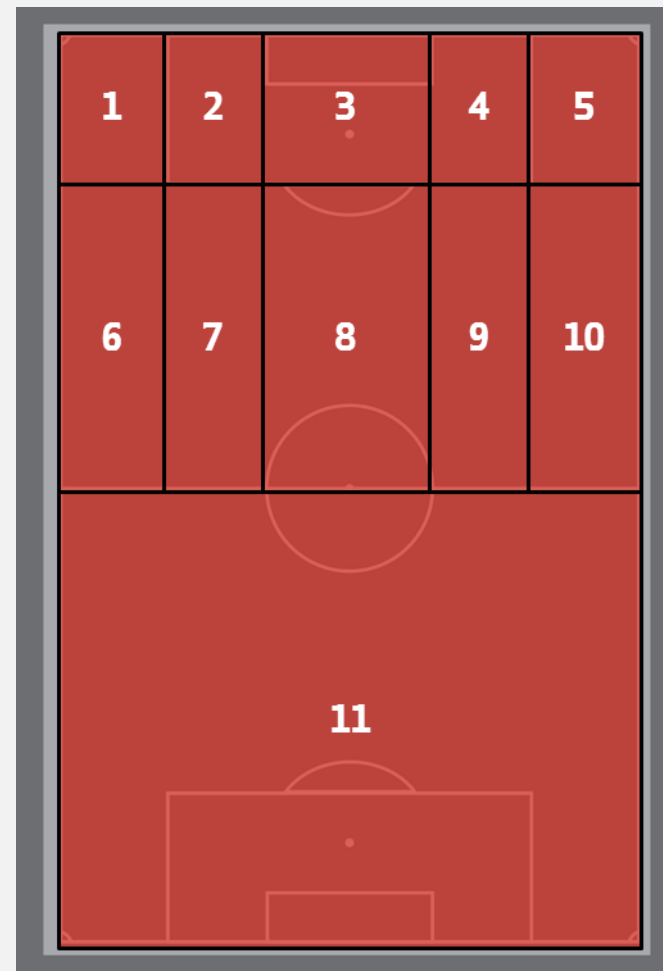


Figure 6: Overview of shot origin zones

Ball control type	a) direct b) volley c) control - shot d) distance covered < 10m e) distance covered > 10m f) set piece taker	<i>Ball control type</i> describes the manner in which the shot taker controls the ball before or during a shot. “Direct” indicates that the shot taker’s first touch of the ball is the shot itself, unless the shot is considered a volley. “Volley” indicates that the shot taker’s first touch of the ball is the shot itself and that the ball has not touched the ground since the last touch by another player or object (post/crossbar...). “Control - shot” indicates that the shot taker controlled the ball with a single touch without changing his position significantly before taking the shot. “Distance covered < 10m” indicates that the shot taker went on a run with the ball, taking more than one touch, and covered a distance of less than 10 metres. “Distance covered > 10m” indicates that the shot taker went on a run with the ball and covered a distance of more than 10 metres. “Set piece taker” indicates that the shot taker is the player taking a set piece.
Prior ball path	a) header b) long pass from open play c) other pass from open play d) one-two e) cross from open play f) shot g) free kick h) corner kick i) throw-in j) teammate action k) ricochet off the goal frame l) ricochet off the goalkeeper m) ricochet off an opposing outfield player n) ricochet off a teammate o) other ricochet p) opponent’s misplaced pass q) opponent’s goalkeeper action r) opponent’s defensive action s) other action by an opponent t) own set piece	<i>Prior ball path</i> describes the manner in which the ball directly reached the shot taker. “Header” indicates that the ball reached the shot taker through a teammate’s header, unless it was a shot, one-two or a cross; “Long pass from open play” indicates that the ball reached the shot taker through a teammate’s long pass from open play. “Other pass from open play” indicates that the ball reached the shot taker through a teammate’s short or medium length pass from open play, unless it was a one-two. “One-two” indicates that the ball reached (or was returned to) the shot taker through a one-two. “Cross from open play” indicates that the ball reached the shot taker through a cross from open play. “Shot” indicates that the ball reached the shot taker through a teammate’s preceding shot. “Free kick” indicates that the ball reached the shot taker through a teammate’s free kick, unless it was a shot. “Corner kick” and “throw-in” indicate that the ball reached the shot taker through a corresponding set piece executed by a teammate. “Teammate action” indicates that the ball reached the shot taker from a teammate, but not through a ricochet or any of the previously mentioned options. “Ricochet” in combination with either “off the goal frame”, “off the goalkeeper”, “off an opposing outfield player”, “off a teammate” or “other” indicates that the ball reached the shot taker through a corresponding ricochet. Ricochet means that an outfield player or goalkeeper was hit by a shot, only touched the ball once and appeared to have been unable to consciously influence or direct the ball’s resulting direction. A ricochet also refers to a shot hitting the post, crossbar, referee or corner flag and rebounding back into play. “Opponent’s misplaced pass”, “opponent’s goalkeeper action”, “opponent’s defensive action” or “other action by an opponent” indicate that the ball reached the shot taker through a corresponding action by an opponent, but not through a ricochet. Defensive actions by outfield players include clearances, other clearance attempts, defensive actions in challenges, etc. Other actions include situations in which the shot taker wins the ball from an opponent dribbling the ball or receives it from an opponents’ set piece.

		“Own set piece” indicates that the shot taker executed a set piece as a shot. If the shot occurs after a free kick layoff, the manifestation free kick is always to be used for the attribute prior ball path, even if the ball is laid off or stopped for a second time after the execution of the free kick. If another player’s last touch of the ball before the shot only slightly alters the ball’s path, this action is disregarded. If the match situation that leads to the shot is a solo run by the shot taker himself, then the manner in which the shot taker received the ball before his solo run is recorded as the prior ball path.
Shot setup action	a) header b) long pass from open play c) other pass from open play d) one-two e) cross from open play f) shot g) free kick h) corner kick i) throw-in j) teammate action k) fouled l) other m) no entry	<i>Shot setup action</i> describes the type of action that was used by the shot setup provider. “Header” indicates that the ball reached the shot taker through a teammate’s header, unless it was a shot, one-two or a cross; “Long pass from open play” indicates that the ball reached the shot taker through a teammate’s long pass from open play. “Other pass from open play” indicates that the ball reached the shot taker through a teammate’s short or medium length pass from open play, unless it was a one-two. “One-two” indicates that the ball reached (or was returned to) the shot taker through a one-two. “Cross from open play” indicates that the ball reached the shot taker through a cross from open play. “Shot” indicates that the ball reached the shot taker through a teammate’s preceding shot. “Free kick” indicates that the ball reached the shot taker through a teammate’s free kick, unless it was a shot. “Corner kick” and “throw-in” indicate that the ball reached the shot taker through a corresponding set piece executed by a teammate. “Teammate action” indicates that the ball reached the shot taker from a teammate, but not through any of the previously mentioned options. “Fouled” indicates that the shot setup provider was fouled and that the shot taker executed the subsequent free kick or penalty kick as a shot. “Other” refers to all other situations. If a shot is taken without a shot setup, the manifestation no entry is used.
From a free kick	a) yes b) no	<i>From a free kick</i> indicates that a shot is taken either directly from the free kick or following a free kick layoff.
Scoring chance evaluation	a) clear-cut chance b) chance c) none	<i>Scoring chance evaluation</i> refers to the quality of the scoring chance. A clear-cut chance is defined as a player either taking a shot or having the opportunity to take a shot from a “very promising position”. A very promising position indicates that • he is in close proximity to the opponents’ goal or that only the opposing goalkeeper remains between him and the goal; • he is not being put under much pressure and has a fairly clear sight of goal; • the angle to the goal is not too tight; • no external circumstances are impeding his ability to control the ball or rendering it impossible (a ball played too far behind the player, a ball played too high or hard to execute a controlled header...).

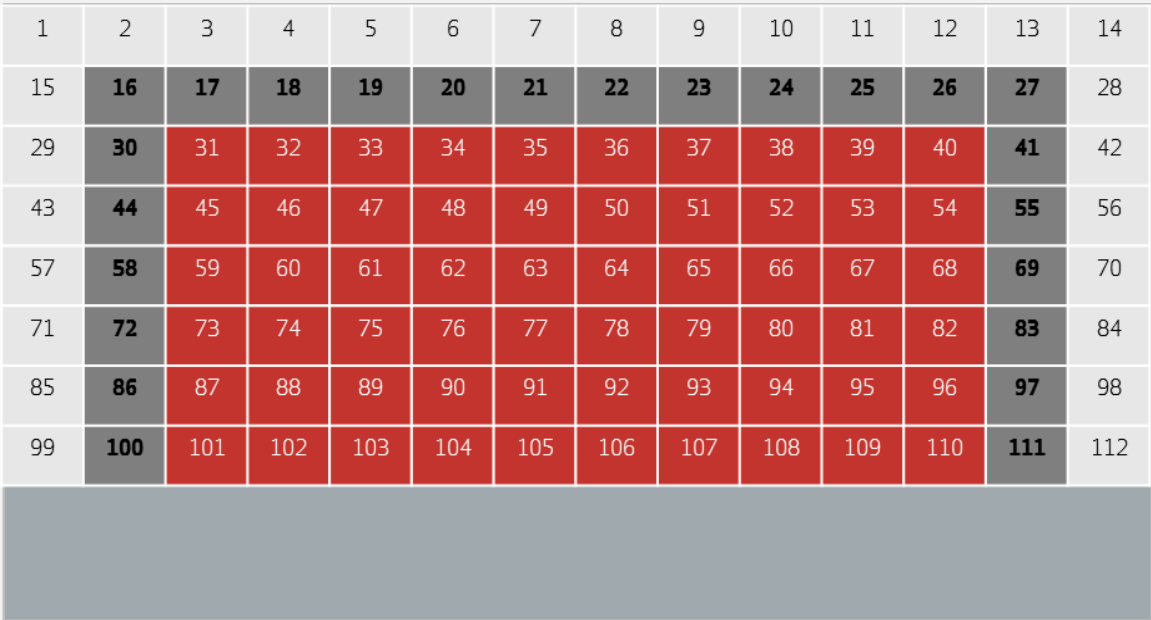
		Therefore, every penalty is a clear-cut chance. Chances are defined as any situation – except clear-cut chances – that create an imminent threat of scoring, meaning there must be a “real chance” of scoring from the situation. This threat of scoring can be created by - the good shooting position of the attacking player; - the particularly good execution of a shot (e.g. when a shot from long distance can only be saved by the goalkeeper with great difficulty); - an (unsuccessful) action by a defending player, which (almost) leads to the ball landing in his own net. If a goal or own goal is indeed scored during the match, the situation may be deemed a clear-cut chance. It is always at least deemed a chance, however.
Player responsible for a clear-cut chance	a) reference person ID b) no entry	<i>Player responsible for a clear-cut chance</i> refers to the player that created the clear-cut chance. This could be the shot taker himself (e.g. when a clear-cut chance is created by an individual run), the shot setup provider or a third player (e.g. when he plays the decisive ball to put a teammate through on goal, who then squares the ball to another teammate).
Shot circumstances	a) not impeded b) impeded c) highly impeded	<i>Shot circumstances</i> indicates whether influences impeded the execution of a shot. In this case it is not the quality and execution type of the shot that should be considered, but the circumstances under which the shot was taken. “Not impeded” means that the shot was not influenced by any negative factors. “Impeded” indicates that the prior ball path hampered the shot taker’s execution. The reasons for this may include the following: - it was difficult for the shot taker to reach or to deal with the ball - the shot taker had little time to react before taking the shot - the shot taker’s body position or posture had an adverse effect on the shot In relation to this, “highly impeded” indicates that one of the following factors influenced the shot taker: - if was very difficult for the shot taker to reach or to deal with the ball - the shot taker had very little time to react before taking the shot - the shot taker’s body position or posture had an exceptionally adverse effect on the shot
Significance evaluation	a) spectacular b) noteworthy c) relevant d) none e) no entry	<i>Significance evaluation</i> divides the media importance of shots into four quality categories. Successful shots are excluded from this evaluation. Shots classified as “spectacular” are especially high-quality chances, e.g. a brilliant shot off the crossbar, a huge save by the goalkeeper, the best scoring chances. The difference to clear-cut chances is that a shot from a poor position can be

		classified as “spectacular” but not as a clear-cut chance. Conversely, a shot during a clear-cut chance isn’t necessarily classified as “spectacular”. Hence, the significance evaluation is based more on the effect of the shot, whilst the manifestation clear-cut chance is based on the initial situation before the shot. Shots classified as “noteworthy” are good and remarkable scoring opportunities – a big scoring chance, a dangerous finish. Shots classified as “relevant” are scoring chances that would be counted towards a “ratio of chances” and that carry a relevant scoring threat. Shots classified as “none” carry no real scoring threat and would not be considered real “scoring chances” – a shot quickly blocked, a glancing header, etc.
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Working Version

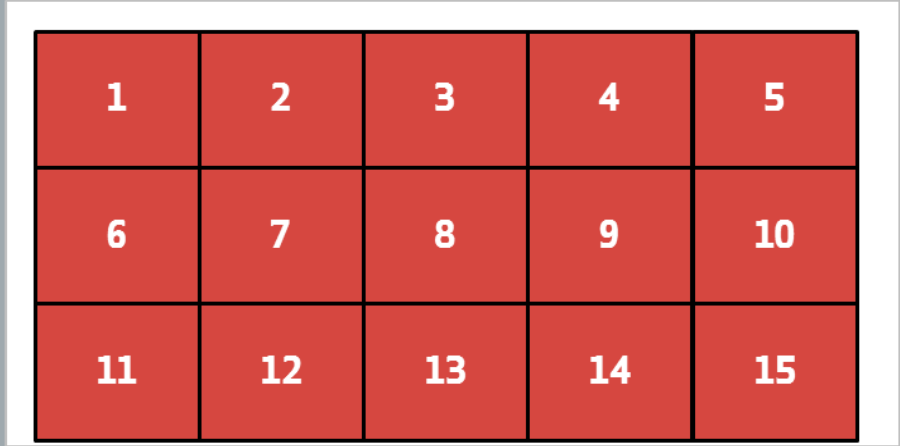
Counterattack	a) yes b) before a penalty kick c) no	<i>Counterattack</i> indicates whether a counterattack preceded a shot. Counterattack refers to a match event during which a team gains ball control in its own half, immediately starts a quick counterattack and, depending on where the ball was previously won, takes a shot within 12 to 14 seconds. The match may also be halted during this period of time. If the referee decides to award a penalty kick, taking the above definitions into account, this event is also recorded. Should a penalty kick be followed by one or more follow-up shots, the attribute counterattack with the manifestation before a penalty kick is recorded for all shots during the match situation penalty kick.
Shot situation	a) cross from open play b) pass from open play c) solo run d) free kick e) penalty kick f) corner kick g) throw-in h) opponents' loss of possession	<i>Shot situation</i> refers to a match situation that led to a shot. A distinction is made between the following types of situations: • cross from open play; • pass from open play; • solo run; • set piece: penalty kick, free kick, corner kick, throw-in; • opponents' loss of possession. Only one of these match situations is assigned to every shot. If one of the following actions leads to a shot, the respective action is treated as a pass: a kick-out or throw-out by the goalkeeper, a clearing ball or similar action, a goal kick, kick-off or dropped ball. Crosses, passes, solo runs and set pieces must always be executed by the team that then takes the shot. Conversely, it is the other way around for the manifestation opponents' loss of possession. Opponents' loss of possession refers to a match situation during which the team in certain ball control loses possession during build-up play (e.g. through a misplaced pass or through a challenge lost in ball control). It is not considered an opponents' loss of possession if, during an attack by the opposing team, the ball is immediately returned to that team due to a defensive attempt without certain ball control (e.g. through a clearance, a punch by the goalkeeper or a successful tackle), as the opponents can continue or resume their attacking move. Finishes (such as shots) or other actions that aim to create an immediate scoring threat (attempting to break through the defence, crosses, passes into the opponents' penalty area...) are not considered to be build-up play actions. Therefore, if possession changes after such an action, this is also not considered an opponents' loss of possession. If the shot taker receives the ball directly from an opponent's misplaced pass, the shot situation is not classified as "pass from open play", but as "opponents' loss of possession". Conversely, whilst a specific match situation is assigned to every shot, a single match situation can also lead to multiple shots. For example, if a cross leads to a shot and a subsequent follow-up shot, the cross is considered the applicable match situation for both shots. As a rule, when choosing which match situation to assign to a shot, the last situation that occurred is to be assigned. However, the following exceptions apply:

		The ball is played into the opponents' penalty area from a cross, corner kick, throw-in or free kick or is already inside the opponents' penalty area at a penalty kick. A shot is taken either before the ball leaves the penalty area or directly after being received by the shot taker from within the penalty area. In this case, the match situation assigned to the shot is the cross, corner kick, throw-in, free kick or penalty kick, even if the attacking team played one or more passes (inside the penalty area or out of the penalty area) between this action and the shot. A shot occurs after a free kick. One or more planned passes or follow-up actions caused by the execution of the free kick occur between the execution of the free kick and the shot (e.g. when a free kick from the training ground includes a rehearsed sequence of passes leading up to the shot or when a deep free kick is headed back towards goal to set up a shot). In this case, the match situation assigned to the shot is the free kick, even if the attacking team played one or more passes between the free kick and the shot. During a sequence of two passes, the first pass is directly knocked on or laid off over a short distance. In this case, the match situation assigned to the shot is the first of the two passes.
Shot setup position	a) left b) half-left c) half-right d) right	<i>Shot setup position</i> describes the position that the shot setup occurred in, from the point of view of the team taking the shot. In the case of a penalty kick, the position of the situation that the penalty kick originated from is recorded, not the position of the penalty kick itself.
Shot tactics (only for penalty kicks)	a) placed shot b) blasted shot c) dummy d) no entry	The attribute <i>shot tactics</i> describes the manner in which the penalty taker executes the shot at the penalty kick.
Target zone at penalty kick	a) numeric b) no entry	<i>Target zone at penalty kick</i> indicates the zone which the ball would pass through once it reaches the goal line, if it wasn't deflected by the opposing goalkeeper.

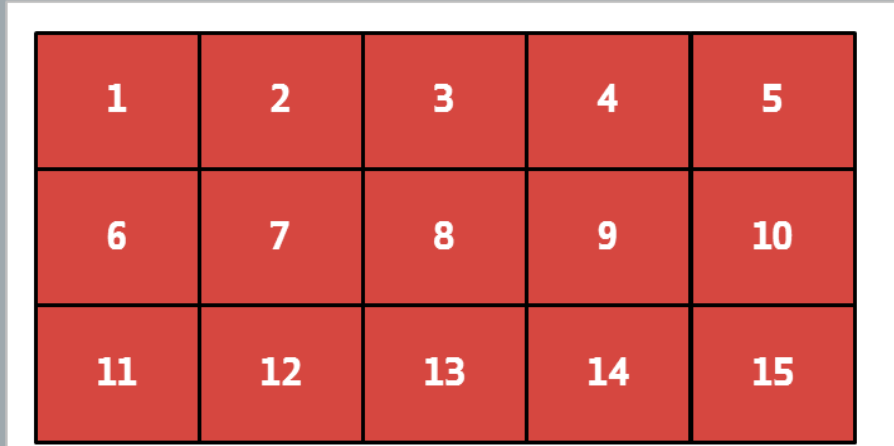
		 Figure 7: Overview of target zones at a penalty kick from the point of view of the attacking team
Shot intention (direct free kick)	a) goalkeeper's corner b) corner behind the wall c) under the wall d) secondary plan of action e) other f) no entry	<i>Shot intention</i> refers to the space which is the intended target of a player executing a direct free kick as a shot. If the free kick is only executed as a shot due to the player's secondary tactical plan of action, the manifestation secondary plan of action is used.
Rotation	a) inswinging b) outswinging c) other d) no entry	The <i>rotation</i> attribute is only recorded for a shot if a direct free kick is executed as a shot. The manifestation other is to be used in this case. The manifestation no entry is to be used for all other shots.
Successful shot		<i>Successful shot</i> refers to a shot, which leads to the referee awarding a goal in accordance with the DFB football rules. The successful shot is usually awarded to the player that took the shot. Should a teammate or opposing player perform

		a ball action before the ball crosses the goal line, the goal (or own goal) is awarded to that player instead. However, it is essential that the touch of the ball can be clearly identified as an active action. In the context of this action, this is also the case for rule violations that have not been penalised by the referee (e.g. an offside position or a handball). The first shot is then recorded as a shot blocked. If a touch of the ball is performed without ball control before the ball crosses the goal line, this is disregarded. The goal is awarded to the last active player. This also applies to the following situations: • The shot ricochets off the post or crossbar and a subsequent touch of the ball without ball control by another player leads to a goal. • A ball is played towards goal and a subsequent touch of the ball without ball control by another player leads to a goal. In this case, it is assumed that the initial player had the secondary plan of action of taking a shot (and the ball played is recorded as a shot). There is one exception: if a ball is clearly played away from goal and a subsequent touch of the ball without ball control leads to a goal, the goal is awarded to the player that touched the ball without ball control. If a player of the team in ball control is the only player to touch the ball after an indirect free kick or throw-in was taken and the ball then ends up in the goal, the goal is awarded to the player that last touched the ball. Regardless of the rules laid out above, the DFL is responsible for the final decision as to which player is awarded the goal.
Assist player	a) reference person ID b) no entry	<i>Assist player</i> refers to the person that provided the assist for a goal. This is the last player of the scoring team to touch the ball before a successful shot is recorded. The following exceptions apply for a touch of the ball • after which the opposing team has certain ball control before the goalscorer receives it. If a player clears the ball, he is only deemed to have had certain ball control if he had the option of playing the ball to a teammate instead of clearing it. • after which the match is halted by the referee before the goalscorer receives the ball (e.g. at penalty kicks, free kicks and corner kicks directly executed as shots). In these situations, and in all other situations where the assist provider would also be considered the goalscorer, the manifestation no entry is used for both the attribute player as well as the attribute assist type.
Assist type	a) direct b) indirect c) other d) free kick e) no entry	<i>Assist type</i> refers to the type of assist provided. A distinction is made between the following cases: A direct assist indicates that a ball action is performed just (usually a few seconds) before the match event successful shot (goal), with the goalscorer receiving the ball. Touches of the ball by opposing players after this ball action are disregarded, provided the goalscorer would presumably have also gained ball control in a comparable match situation without these touches occurring. An indirect assist refers to a ball action, after which the direction of the ball is significantly altered by an opposing player before reaching the goalscorer, provided the goalscorer would presumably not have gained ball control in a comparable match situation without this ball action occurring. An indirect assist also refers to a situation during which the ball comes off the goal frame, is blocked by an opponent or is saved by the goalkeeper, before reaching the goalscorer. If, in this case, the goalscorer and the player that took the first shot are one and the same, the indirect assist is awarded to the player who performed the last ball action before the goalscorer (usually the player that played the final ball before the first shot).

		Other indicates that the ball was played directly to the shot taker by a teammate, whose primary intention was not to provide an assist (e.g. a pass to the goalscorer in the team's own half, after which the goalscorer goes on an individual run with the ball before scoring; a player is hit by a shot without becoming active; a teammate's shot directly reaches the goalscorer). A free kick assist indicates that the goalscorer receives the ball from a free kick layoff at a free kick. If the goal is scored from a follow-up shot after a penalty kick or a free kick executed as a shot, the following rules apply: If the goalscorer is the player that won or took the penalty kick or free kick, he is awarded a fouled player assist, provided the ball was not touched by any of the goalscorer's teammates before the follow-up shot (e.g. if a teammate in the wall is hit by the initial shot). If the goalscorer is not the player that took the penalty kick or free kick, then the player that took the initial shot from the set piece is awarded an indirect assist, provided the ball was not touched by any of the goalscorer's teammates before the follow-up shot.
Fouled player assist	a) reference person ID b) no entry	<i>Fouled player assist</i> refers to when a goal is scored from a penalty kick or free kick, which was awarded due to a foul against an opposing player. This applies to free kicks that are directly executed as shots or when a free kick layoff leads to a shot. In these cases, the attribute fouled player assist refers to the player that was fouled to win the set piece in question. If the manifestation no entry has been used for the attribute assist player, then the fouled player can be deemed to have provided the assist (depending on the understanding of what constitutes an assist). This is also the case if the manifestation free kick has been used for the attribute assist type.
Goal zone	a) numeric	<i>Goal zone</i> refers to the position at which the ball crosses the goal line, from the point of view of the goalscorer.

		 Figure 8: Overview of goal zones from the point of view of the attacking team
Individual action	a) yes b) no	<i>Individual action</i> indicates that an individual action was performed by the goalscorer before the successful shot. An individual action is defined as a player's remarkable individual contribution
Deflecting outfield player	a) reference person ID b) no entry	<i>Deflecting outfield player</i> refers to the person that last touched the ball before it crossed the goal line, other than the goalscorer.
Deflecting goalkeeper	a) reference person ID b) no entry	<i>Deflecting goalkeeper</i> indicates whether the goalkeeper of the team without ball control touched the ball before it crossed the goal line.
Contribution to a goal	a) reference person ID b) no entry	<i>Contribution to a goal</i> is awarded to a player of the scoring team, if he is not the assist provider, but has made a significant contribution to setting up the goal through his action. In particular, a contribution to a goal is awarded for the following actions:

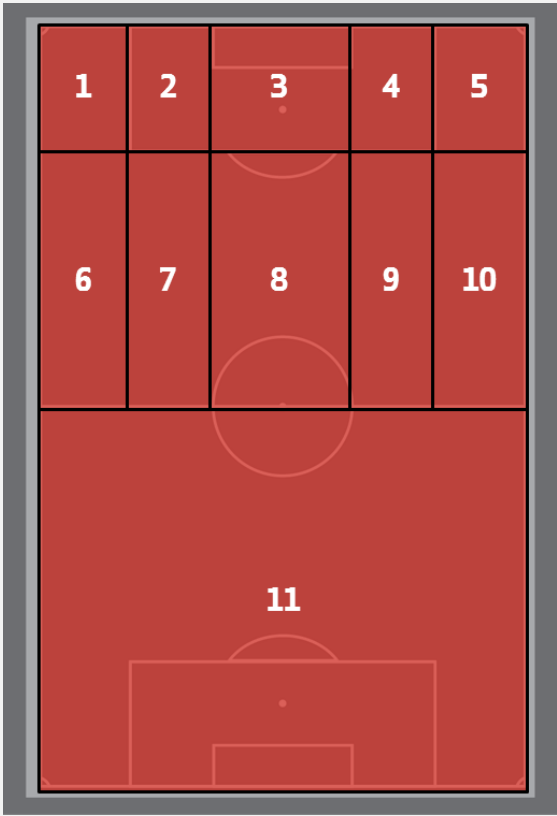
		- the player had possession or a ball action before the assist provider and significantly contributed to the creation of the successful shot; - the player let the ball through for the goalscorer without touching the ball himself, thereby significantly contributing to the creation of the successful shot. A contribution to a goal is only awarded for the execution of a set piece if the type and quality of the execution clearly contributed to the scoring of the goal. Per goal, only one person may be awarded a contribution to the goal. A player could also be awarded a contribution to a goal that he himself scored, but only if a teammate's (assist provider's) action occurred between his contributing action and the scoring of the goal.
Current score	a) alphanumeric	<i>Current score</i> returns the current score in the match after the successful shot (goal).
Shot saved		<i>Shot saved</i> indicates that the goalkeeper of the team without ball control prevented a shot from becoming a goal by touching the ball. It is also used if the goalkeeper's behaviour indicates that he believes his action will prevent a goal. In addition, the match events ball won, other ball action (clearance) or pass should be recorded, with a reference to the goalkeeper as the player involved. The additional match event depends on the save result.
Shot saved goalkeeper	a) reference person ID	<i>Shot saved goalkeeper</i> refers to the goalkeeper of the team without ball control that saved the shot.
Save type	a) arms b) legs c) other	<i>Save type</i> indicates the body part that the goalkeeper of the team without ball control used to save the shot.
Save result	a) held b) cleared c) not cleared	<i>Save result</i> indicates whether the ball was held by the goalkeeper (held), not held by the goalkeeper with no scoring threat remaining (cleared) or with a scoring threat remaining after the save was made (not cleared).
Touched onto the goal frame	a) left post b) right post c) crossbar d) no entry	<i>Touched onto the goal frame</i> indicates whether the goalkeeper's touch diverted the ball onto the goal frame. Left/right post refers to the goalposts from the point of view of the attacking team.
Goalkeeper's save evaluation	a) extraordinary save b) normal save c) gathered without difficulty	<i>Goalkeeper's save evaluation</i> refers to the goalkeeper's performance during a shot saved. "Extraordinary save" indicates that the goalkeeper saved a shot by the opposing team that was very difficult to save. Only the level of difficulty is of relevance, not whether the goalkeeper's action was particularly beautiful or spectacular. If a goalkeeper saves a shot by the opposing team which is also classified as a clear-cut chance, this does not necessarily mean that he made an "extraordinary save" (e.g. if the opponent's shot is quite poor and not very

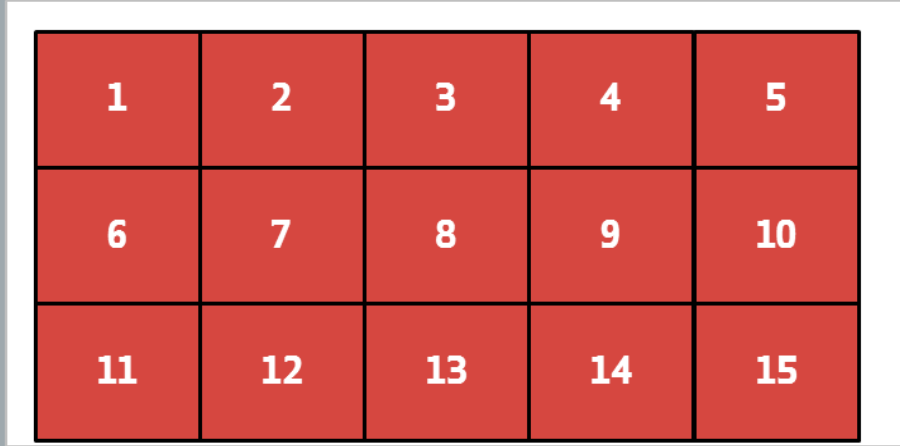
		difficult for the goalkeeper to save). A saved penalty is classified as an “extraordinary save”, whilst “normal save” refers to shots saved that are not recorded in one of the other two categories. “Gathered without difficulty” refers to saves that are very easy for the goalkeeper to make, e.g. merely picking a ball up off the ground or saving a poor shot from distance.
Goal zone	a) numeric b) no entry	<i>Goal zone</i> refers to the position at which the ball would have crossed the goal line, from the point of view of the goalscorer. It is only to be used in the case of a saved penalty kick.  Figure 8: Overview of goal zones from the point of view of the attacking team
Shot blocked		<i>Shot blocked</i> indicates that after a shot was taken, at least one player who is not the goalkeeper prevented a goal or prevented the ball from travelling towards goal as intended, through a touch of the ball. A shot blocked means that the ball is either blocked completely or that the ball’s direction and/or velocity are significantly altered. If the ball is only slightly deflected, the shot should be recorded according to its further course (e.g. shot saved or shot off the goal frame). When distinguishing between a shot saved and a shot blocked, the action that contributed more to preventing the goal is considered. If an opposing player’s shot is deflected in such a way, that the referee subsequently awards a throw-in, the shot is recorded as a shot blocked, provided the player that made the block is not the goalkeeper. If an

		opposing player's shot is deflected in such a way, that the referee subsequently awards a corner kick, the shot is only recorded as a shot blocked if no immediate scoring threat arises after the deflection and if the player that made the block is not the goalkeeper. In all other cases a shot wide is recorded. If a shot is deflected by an attacking team's own player, the same rules apply for choosing the corresponding shot type. If a shot is deflected by the goalkeeper outside the penalty area, then the goalkeeper is to be treated as an outfield player. In this case, the shot is not recorded as a shot saved, but as a shot blocked.
Blocking player	a) reference person ID b) no entry	<i>Blocking player</i> refers to the person blocking the shot. If the shot is blocked by players in the wall and the blocking player cannot be identified beyond doubt, the manifestation no entry is used. All blocking players are awarded an additional match event. The following rules apply: If the blocking player is not capable of implementing a tactical plan of action (e.g. because he does not have enough time to react), the action is recorded as the attribute other ball action. If the blocking player actively reacts to the inbound shot without the ball remaining in his possession, this is recorded as the attribute other ball action and the manifestation clearance. If a player of the team without ball control blocks a shot and subsequently remains in possession (e.g. if the ball gets stuck between the legs of the blocking player and he then takes it under control), the following action performed by the blocking player (pass, challenge, etc.) will decide which event is to be recorded. In addition, the blocking player is recorded as having won the ball. If a blocking player of the attacking team actively reacts to an inbound shot, the following action performed by the blocking player (shot, pass, etc.) will decide which event is to be recorded. For example, if the ball is deflected in such a way that it continues to move towards goal, this action is considered to a shot.
Goal prevented	a) yes b) no	<i>Goal prevented</i> indicates whether a player of the team without ball control prevented a <u>certain</u> goal by blocking the shot.
Touched onto the goal frame	a) left post b) right post c) crossbar d) no entry	<i>Touched onto the goal frame</i> indicates whether a player of the team without ball control, who is not the goalkeeper, diverted the ball onto the goal frame by significantly deflecting the ball. Left/right post refers to the goalposts from the point of view of the attacking team.
Blocked by own team	a) yes b) no entry	<i>Blocked by own team</i> indicates whether one of the shot taker's own teammates blocked the shot or whether it was a defending player.
Shot off the goal frame		<i>Shot off the goal frame</i> refers to a shot that hits the crossbar or one of the posts and for which the referee does not award a successful shot (goal). If the ball is only slightly deflected after the shot has been taken, the shot is also to be recorded as a shot off the goal frame and not as a shot blocked. Whether the goalkeeper of the team without ball control touches the ball after it has hit the post or the crossbar is irrelevant and does not result in the shot being recorded as a shot saved, unless it is clear that a successful shot would have occurred without the goalkeeper's immediate ball action. In this case, a shot saved is to be recorded, not a shot off the goal frame.

Area of impact	a) left post b) right post c) crossbar	<i>Area of impact</i> refers to the part of the goal frame that the ball hits, from the point of view of the executing team's direction of play.
Shot wide		<i>Shot wide</i> refers to a shot that misses the goal to the right, left or over the top as it crosses the goal line.
Placement	a) close b) far	<i>Placement</i> indicates whether the shot missed the goal by more or less than two metres.
Crossing goal line or touchline	a) left of goal b) right of goal c) straight over the top of goal d) over the right of goal e) over the left of goal f) over the touchline	<i>Crossing goal line or touchline</i> indicates whether the shot missed the goal on the right, left or over the top or whether it crossed the touchline, from the point of view of the executing team's direction of play. If the ball misses the goal to either of the sides and also goes over the top of goal, the manifestations over the right of goal or over the left of goal are used. Only shots that miss the goal over the top and cross in between the (extended) goal posts are recorded as having gone straight over the top of goal.
Other shot		<i>Other shot</i> refers to a match event which is not classified as a successful shot, shot saved, shot blocked, shot off the goal frame or shot wide. Other shot is recorded in the following situations, for example: A player does not strike the ball cleanly, which results in the ball coming to rest shortly after the player's touch. This situation is distinguished from other ball action due to the player's obvious intention to execute a shot and his playing the ball with the intended body part. A shot is intercepted well wide of the goal before it has crossed the goal line.
Own goal		<i>Own goal</i> is a ball action during which a player unsuccessfully attempts to implement his tactical plan of action, after which the referee awards a goal to the opposing team. A match event of the own goal type occurs if the attacking team does not take a shot during the immediately preceding match event, a defending player touches the ball, which ends up in his own goal and the referee then awards a goal in accordance with the DFB football rules. The important aspect is whether the defending player is deemed to have actively implemented a tactical plan of action, i.e. a ball action, and whether this action was unsuccessful. If this is the case, an own goal should be recorded. However, under certain circumstances a ball action classified as other ball action can also lead to an own goal. An own goal is also recorded if a defending player is the only player to touch the ball after the execution of an indirect free kick or a throw-in and the ball then ends up in his own goal, after which the referee awards a goal in accordance with the DFB football rules. In this case, the player in question is recorded as the scorer of the own goal. If several defending players touch the ball before it ends up in their goal, the player that last performed a ball action, i.e. was capable of implementing a tactical plan of action, before the ball crosses the goal line is recorded as the scorer of the own goal. For example, if a player/goalkeeper only touches the ball in his attempt to prevent a teammate's own goal and he is unable to do so, he will not be recorded as the scorer of the own goal; the teammate whose own goal he was trying to prevent will be recorded instead (e.g. a defender deflects a ball towards his own goal in an attempt to

		defend a cross, after which the goalkeeper tries to make a save and touches the ball before it ends up in the goal). There is one exception to this rule. If the ball is clearly played away from goal by a defending player before hitting a teammate, which results in the ball ending up in their goal, then the player that last touched the ball is recorded as the scorer of the own goal. Regardless of the rules laid out above, the DFL is responsible for the final decision as to which player is awarded the goal.
Team	a) reference club ID	<i>Team</i> refers to the team of the player that scored the own goal.
Player	a) reference person ID	<i>Player</i> refers to the person that scored the own goal.
Own goal in favour of team	a) reference club ID	<i>Own goal in favour of team</i> refers to the team that did not score the own goal.
Phase of possession number	a) numeric	<i>Phase of possession number</i> returns the phase of possession in question.
Own goal after counterattack	a) yes b) no	<i>Own goal after counterattack</i> indicates whether the own goal was preceded by a counterattack by the team in whose favour the own goal is awarded. In this case, counterattack refers to a match event during which a team gains ball control in its own half, immediately starts a quick counterattack after which, depending on where the ball was previously won, an own goal is awarded within 12 to 14 seconds. The match may also be halted during this period of time.
Shot type	a) left leg b) right leg c) head d) torso	<i>Shot type</i> indicates whether the own goal was scored with the left leg or right leg, or through the use of the head or torso. If the own goal is scored through the use of the arm or hand, then the shot type torso is to be used.
Shot origin	a) numeric	<i>Shot origin</i> refers to where the scorer of the own goal is located at the time of scoring, from the point of view of the team in whose favour the own goal is awarded.

		 Figure 6: Overview of shot origin zones
Goal zone	a) numeric	<i>Goal zone</i> refers to the position at which the ball crosses the goal line, from the point of view of the scorer of the own goal.

		 Figure 8: Overview of goal zones from the point of view of the attacking team
Ball control type	a) direct b) volley c) control - shot d) distance covered < 10m e) distance covered > 10m	<i>Ball control type</i> describes the manner in which the scorer of the own goal controls the ball before or during the scoring of the own goal. “Direct” indicates that the scorer’s first touch led to the scoring of the own goal. “Volley” indicates that the scorer’s first touch led to the scoring of the own goal and that the ball has not touched the ground since the last touch by another player or object (post/crossbar...). “Control - shot” indicates that the scorer controlled the ball with a single touch without changing his position significantly before scoring the own goal. “Distance covered < 10m” indicates that the scorer went on a run with the ball, taking more than one touch, and covered a distance of less than 10 metres before scoring the own goal. “Distance covered > 10m” indicates that the scorer went on a run with the ball and covered a distance of more than 10 metres before scoring the own goal.

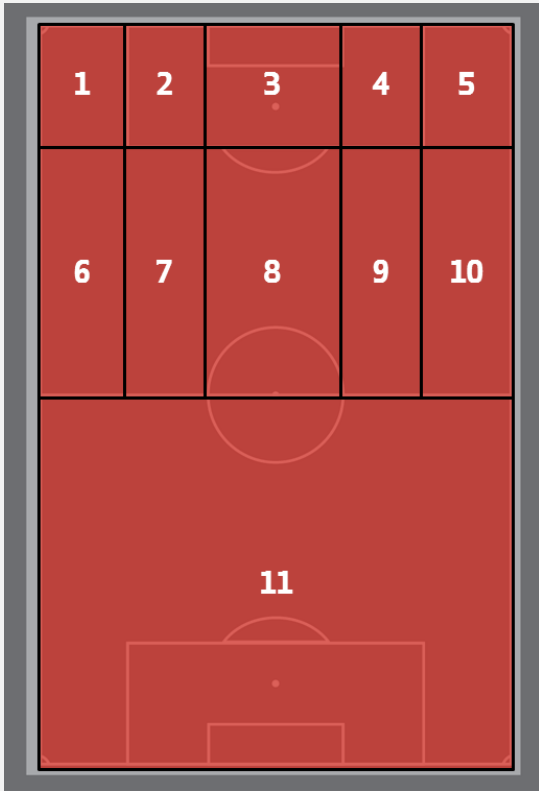
Prior ball path	a) header b) long pass from open play c) other pass from open play d) cross from open play e) shot f) free kick g) corner kick h) throw-in i) teammate action j) ricochet off the goal frame k) ricochet off the goalkeeper l) ricochet off an opposing outfield player m) ricochet off a teammate n) other ricochet o) opponent's misplaced pass p) opponent's goalkeeper action q) opponent's defensive action r) other action by an opponent	<i>Prior ball path</i> describes the manner in which the ball directly reached the scorer of the own goal. When deciding which manifestation should be used, players of the team in whose favour the own goal is awarded are referred to as teammates, whilst players on the own goal scorer's team are referred to as opposing players or opponents. "Header" indicates that the ball reached the scorer of the own goal through a teammate's header, unless it was a shot or a cross; "Long pass from open play" indicates that the ball reached the scorer of the own goal through a teammate's long pass from open play. "Other pass from open play" indicates that the ball reached the scorer of the own goal through a teammate's short or medium length pass from open play. "Cross from open play" indicates that the ball reached the scorer of the own goal through a teammate's cross from open play. "Shot" indicates that the ball reached the scorer of the own goal through a teammate's preceding shot. "Free kick" indicates that the ball reached the scorer of the own goal through a teammate's free kick, unless it was a shot. "Corner kick" and "throw-in" indicate that the ball reached the scorer of the own goal through a corresponding set piece executed by a teammate. "Teammate action" indicates that the ball reached the scorer of the own goal from a teammate, but not through a ricochet or any of the previously mentioned options. "Ricochet" in combination with either "off the goal frame", "off the goalkeeper", "off an opposing outfield player", "off a teammate" or "other" indicates that the ball reached the scorer of the own goal through a corresponding ricochet. Ricochet means that an outfield player or goalkeeper was hit by a shot, only touched the ball once and appeared to have been unable to consciously influence or direct the ball's resulting direction. A ricochet also refers to a shot hitting the post, crossbar, referee or corner flag and rebounding back into play. "Opponent's misplaced pass", "opponent's goalkeeper action", "opponent's defensive action" or "other action by an opponent" indicate that the ball reached the scorer of the own goal through a corresponding action by an opponent, but not through a ricochet. Defensive actions by outfield players include clearances, other clearance attempts, defensive actions in challenges, etc. If the shot occurs after a free kick layoff, the manifestation free kick is always to be used for the attribute prior ball path, even if the ball is laid off or stopped for a second time after the execution of the free kick. If another player's last touch of the ball before the own goal was scored only slightly alters the ball's path, this action is disregarded. If the match situation that leads to the shot is a solo run by the scorer of the own goal himself, then the manner in which the scorer of the own goal received the ball before his solo run is recorded as the prior ball path.
Own goal situation	a) cross from open play b) pass from open play c) solo run d) free kick e) penalty kick f) corner kick	<i>Own goal situation</i> refers to a match situation that led to an own goal. A distinction is made between the following types of situations: • cross from open play; • pass from open play; • solo run;

	g) throw-in h) own loss of possession	• set piece: penalty kick, free kick, corner kick, throw-in; • own loss of possession. Only one of these match situations is assigned to every own goal. If one of the following actions leads to an own goal, the respective action is treated as a pass: a kick-out or throw-out by the goalkeeper, a clearing ball or similar action, a goal kick, kick-off or dropped ball. Crosses, passes, solo runs and set pieces must always be executed by the team in whose favour the own goal is awarded. Conversely, it is the other way around for the manifestation own loss of possession. As a rule, when choosing which match situation to assign to an own goal, the last situation that occurred is to be assigned. However, the following exceptions apply: The ball is played into the team's own penalty area from a cross, corner kick, throw-in or free kick or is already inside the team's own penalty area at a penalty kick. An own goal is scored either before the ball leaves the penalty area or directly after being received by the scorer of the own goal from within the penalty area. In this case, the match situation assigned to the own goal is the cross, corner kick, throw-in, free kick or penalty kick, even if the attacking team played one or more passes (inside the penalty area or out of the penalty area) between this action and the own goal. An own goal is scored after a free kick. One or more planned passes or follow-up actions caused by the execution of the free kick occur between the execution of the free kick and the own goal (e.g. when a free kick from the training ground includes a rehearsed sequence of passes leading up to the shot or when a deep free kick is headed back towards goal to set up a shot). In this case, the match situation assigned to the own goal is the free kick, even if the attacking team played one or more passes between the free kick and the shot. During a sequence of two passes, the first pass is directly knocked on or laid off over a short distance. In this case, the match situation assigned to the own goal is the first of the two passes.
Shooting setup position	a) left b) half-left c) half-right d) right	<i>Shooting setup position</i> describes the position of the own goal situation from the point of view of team in whose favour the own goal is awarded.
Current score	d) alphanumeric	<i>Current score</i> returns the current score in the match after the own goal.

Own goal prevented		<i>Own goal prevented</i> is a player action during which a player of the team that would have scored an own goal prevents the match event own goal from occurring. An action is to be recorded as own goal prevented if a defending player plays the ball towards his own goal in such a way that one of his teammates must make a significant intervention to prevent the match event own goal from occurring. If the player in question is the defending goalkeeper, then “significant” means that the goalkeeper’s action must constitute a noteworthy play (e.g. saving a slow back pass played straight at the goalkeeper would not constitute a noteworthy play). If an own goal is prevented, both the player that would have scored the own goal as well as the player that prevented the own goal are each assigned a match event other ball action or ball played. This decision should be based on the action that took place.
Team	a) reference club ID	<i>Team</i> refers to the team that prevented the own goal.
Player	a) reference person ID	<i>Player</i> refers to the person that prevented the own goal.
Save type	a) arms b) legs c) other d) no entry	<i>Save type</i> indicates the body part that the goalkeeper used to prevent an own goal from being scored. If an outfield player prevents an own goal, the manifestation no entry is used for the attribute save type.
Save result	a) held b) cleared c) not cleared d) no entry	<i>Save result</i> describes the manner in which the own goal was prevented, if it was prevented by the goalkeeper of the team that would have scored the own goal. The attribute indicates whether the ball was held by the goalkeeper (held), not held by the goalkeeper with no scoring threat remaining (cleared) or with a scoring threat remaining after the save was made (not cleared). If an outfield player prevents an own goal, the manifestation no entry is used for the attribute save result.
Touched onto the goal frame	a) left post b) right post c) crossbar d) no entry	<i>Touched onto the goal frame</i> indicates whether the player’s touch diverted the ball onto the goal frame in his attempt to prevent an own goal from being scored. Left/right post refers to the goalposts from the point of view of the attacking team.
Loss of possession before goal conceded		<i>Loss of possession before goal conceded</i> indicates whether a successful shot was taken within approx. 12 to 14 seconds after an opponents’ loss of possession, regardless of whether the loss of possession is the match situation assigned to the shot. The game may not have been halted within this period of time. Opponents’ loss of possession refers to a match situation during which the team in certain ball control loses possession during build-up play (e.g. through a misplaced pass or through a challenge lost in ball control). It is not considered an opponents’ loss of possession if, during an attack by the opposing team, the ball is immediately returned to that team due to a defensive attempt without certain ball control (e.g. through a clearance, a punch by the goalkeeper or a successful tackle), as the opponents can continue or resume their attacking move.

		Finishes (such as shots) or other actions that aim to create an immediate scoring threat (attempting to break through the defence, crosses, passes into the opponents' penalty area...) are not considered to be build-up play actions. Therefore, if possession changes after such an action, this is also not considered an opponents' loss of possession.
Team	a) reference club ID	<i>Team</i> refers to the team which lost possession before conceding a goal.
Player	a) reference person ID	<i>Player</i> refers to the person that caused the loss of possession before goal conceded. This is the player that last had or could have gained certain ball control.
Loss of possession origin	a) numeric	<i>Loss of possession origin</i> refers to the position where possession was lost, from the point of view of the team who is awarded the goal.

Working Version

		 Figure 6: Overview of loss of possession zones before a goal conceded
Loss of possession type	a) challenge lost in ball control b) misplaced pass c) other	<i>Loss of possession type</i> indicates whether the loss of possession occurred due to a challenge lost in ball control, a misplaced pass or for some other reason.
Chance without a resulting shot		<i>Chance without a resulting shot</i> refers to all match situations during which an immediate scoring threat is created, meaning a real possibility of a goal being scored in this match situation must be given, without a shot being taken or an own goal being scored. If, during a chance without a resulting shot, a player's match action keeps the immediate scoring threat alive, his action is not recorded as a chance without a resulting shot.

Team	a) reference club ID	<i>Team</i> refers to the team whose goal is not threatened.
Player	a) reference person ID	<i>Player</i> refers to the person in ball control during the chance without a resulting shot or the person threatening his own goal.
Chance prevented by the goalkeeper	a) yes b) no	<i>Chance prevented by the goalkeeper</i> indicates whether the goalkeeper touched the ball and prevented the chance from resulting in a goal.
Clear-cut chance	a) yes b) no	A <i>clear-cut chance</i> is defined as a player having the opportunity to take a shot from a “very promising position” in front of the opponents’ goal. A very promising position indicates that • he is in close proximity to the opponents’ goal or that only the opposing goalkeeper remains between him and the goal; • he is not being put under much pressure and has a fairly clear sight of goal; • the angle to the goal is not too tight; • no external circumstances are impeding his ability to control the ball or rendering it impossible (a ball played too far behind the player, a ball played too high or hard to execute a controlled header...). Therefore, every penalty is a clear-cut chance. Chances are defined as any situation – except clear-cut chances – that create an imminent threat of scoring. This threat of scoring can be created by • the good shooting position of the attacking player; • an (unsuccessful) action by a defending player, which (almost) leads to the ball landing in his own net.
Prior ball path	a) header b) long pass from open play c) other pass from open play d) one-two e) cross from open play f) shot g) free kick h) corner kick i) throw-in j) teammate action k) ricochet off the goal frame l) ricochet off the goalkeeper m) ricochet off an opposing	<i>Prior ball path</i> describes the manner in which the ball directly reached the player who had the chance without a resulting shot. If a player threatens his own goal during a chance without a resulting shot, then players of the team in whose favour a goal would have been awarded are referred to as teammates, whilst players whose goal is threatened are referred to as opposing players or opponents. “Header” indicates that the ball reached the player who had the chance through a teammate’s header, unless it was a shot, one-two or a cross; “Long pass from open play” indicates that the ball reached the player who had the chance through a teammate’s long pass from open play. “Other pass from open play” indicates that the ball reached the player who had the chance through a teammate’s short or medium length pass from open play, unless it was a one-two. “One-two” indicates that the ball reached (or was returned to) the player who had the chance through a one-two. “Cross from open play” indicates that the ball reached the player who had the chance through a cross from open play.

	outfield player n) ricochet off a teammate o) other ricochet p) opponent's misplaced pass q) opponent's goalkeeper action r) opponent's defensive action s) other action by an opponent t) own set piece	"Shot" indicates that the ball reached the player who had the chance through a teammate's preceding shot. "Free kick" indicates that the ball reached the player who had the chance through a teammate's free kick, unless it was a shot. "Corner kick" and "throw-in" indicate that the ball reached the player who had the chance through a corresponding set piece executed by a teammate. "Teammate action" indicates that the ball reached the player who had the chance from a teammate, but not through a ricochet or any of the previously mentioned options. "Ricochet" in combination with either "off the goal frame", "off the goalkeeper", "off an opposing outfield player", "off a teammate" or "other" indicates that the ball reached the player who had the chance through a corresponding ricochet. Ricochet means that an outfield player or goalkeeper was hit by a shot, only touched the ball once and appeared to have been unable to consciously influence or direct the ball's resulting direction. A ricochet also refers to a shot hitting the post, crossbar, referee or corner flag and rebounding back into play. "Opponent's misplaced pass", "opponent's goalkeeper action", "opponent's defensive action" or "other action by an opponent" indicate that the ball reached the player who had the chance through a corresponding action by an opponent, but not through a ricochet. Defensive actions by outfield players include clearances, other clearance attempts, defensive actions in challenges, etc. Other actions include situations in which the player who had the chance wins the ball from an opponent dribbling the ball or receives it from an opponents' set piece. "Own set piece" indicates that the player who had the chance executed a set piece as a shot. If the shot occurs after a free kick layoff, the manifestation free kick is always to be used for the attribute prior ball path, even if the ball is laid off or stopped for a second time after the execution of the free kick. If another player's last touch of the ball before the chance only slightly alters the ball's path, this action is disregarded. If the match situation that leads to the shot is a solo run by the player who had the chance, then the manner in which that player received the ball before his solo run is recorded as the prior ball path.
Counterattack	a) yes b) no	<i>Counterattack</i> indicates whether a counterattack preceded the chance. Counterattack refers to a match event during which a team gains ball control in its own half, immediately starts a quick counterattack and, depending on where the ball was previously won, creates a chance without a resulting shot within 12 to 14 seconds. The match may also be halted during this period of time.
Chance situation	a) cross from open play b) pass from open play c) solo run d) free kick e) penalty kick f) corner kick g) throw-in h) direct loss of possession	<i>Chance situation</i> refers to the match situation that led to a chance without a resulting shot. A distinction is made between the following types of situations: • cross from open play; • pass from open play; • solo run; • set piece: penalty kick, free kick, corner kick, throw-in; • direct loss of possession.

		Only one of these match situations is assigned to every chance without a resulting shot. If one of the following actions leads to a chance without a resulting shot, the respective action is treated as a pass: a kick-out or throw-out by the goalkeeper, a clearing ball or similar action, a goal kick, kick-off or dropped ball. Crosses, passes, solo runs and set pieces must always be executed by the team that then has the chance without a resulting shot. Conversely, it is the other way around for the manifestation direct loss of possession. If the player who had the chance without a resulting shot receives the ball directly from an opponent's misplaced pass, the situation is not classified as "pass from open play", but as "direct loss of possession". Conversely, whilst a specific match situation is assigned to every chance without a resulting shot, a single match situation can also lead to multiple chances without resulting shots. As a rule, when choosing which match situation to assign to a chance without a resulting shot, the last situation that occurred is to be assigned. However, the following exceptions apply: The ball is played into the opponents' penalty area from a cross, corner kick, throw-in or free kick or is already inside the opponents' penalty area at a penalty kick. A chance without a resulting shot occurs either before the ball leaves the penalty area or directly after being received from within the penalty area by the player who had the chance without a resulting shot. In this case, the match situation assigned to the chance without a resulting shot is the cross, corner kick, throw-in, free kick or penalty kick, even if the attacking team played one or more passes (inside the penalty area or out of the penalty area) between this action and the chance without a resulting shot. A chance without a resulting shot occurs after a free kick. One or more planned passes or follow-up actions caused by the execution of the free kick occur between the execution of the free kick and the chance without a resulting shot (e.g. when a free kick from the training ground includes a rehearsed sequence of passes leading up to the chance without a resulting shot or when a deep free kick is headed back towards goal to set up a chance without a resulting shot). In this case, the match situation assigned to the chance without a resulting shot is the free kick, even if the attacking team played one or more passes between the free kick and the chance without a resulting shot. During a sequence of two passes, the first pass is directly knocked on or laid off over a short distance. In this case, the match situation assigned to the chance without a resulting shot is the first of the two passes.
Chance setup position	a) left b) half-left c) half-right d) right	<i>Chance setup position</i> describes the position of the chance situation from the point of view of the team whose goal is not under threat.

Chance setup action	a) header b) long pass from open play c) other pass from open play d) one-two e) cross from open play f) shot g) free kick h) corner kick i) throw-in j) teammate action k) other l) no entry	<i>Chance setup action</i> describes the type of action that was used by the player that set up the chance without a resulting shot. “Header” indicates that the ball reached the player who had the chance through a teammate’s header, unless it was a shot, one-two or a cross; “Long pass from open play” indicates that the ball reached the player who had the chance through a teammate’s long pass from open play. “Other pass from open play” indicates that the ball reached the player who had the chance through a teammate’s short or medium length pass from open play, unless it was a one-two. “One-two” indicates that the ball reached (or was returned to) the player who had the chance through a one-two. “Cross from open play” indicates that the ball reached the player who had the chance through a cross from open play. “Shot” indicates that the ball reached the player who had the chance through a teammate’s preceding shot. “Free kick” indicates that the ball reached the player who had the chance through a teammate’s free kick, unless it was a shot. “Corner kick” and “throw-in” indicate that the ball reached the player who had the chance through a corresponding set piece executed by a teammate. “Teammate action” indicates that the ball reached the player who had the chance from a teammate, but not through any of the previously mentioned options. “Other” refers to all other situations. If a chance without a resulting shot occurs without a chance setup, the manifestation no entry is used.
Chance setup player	a) reference person ID b) no entry	<i>Chance setup player</i> refers to the person that set up the chance without a resulting shot. This is the last player of the team that is threatening to score to touch the ball before the chance occurs. The following exceptions apply for a touch of the ball • after which the opposing team has certain ball control before the player who had the chance receives it. If a player clears the ball, he is only deemed to have had certain ball control if he had the option of playing the ball to a teammate instead of clearing it. • after which the match is halted by the referee before the player who had the chance receives the ball (e.g. if the player who had the chance executes a penalty kick or free kick). In these situations, and in all other situations where the player who set up the chance would also be considered the player who had the chance, the manifestation no entry is used for both the attribute player as well as the attribute chance setup type.
Chance setup type	a) direct b) indirect c) other d) free kick e) no entry	<i>Chance setup type</i> refers to the type of chance setup. A distinction is made between the following cases: A direct chance setup is a ball action executed within a short period of time (usually a few seconds) before the chance without a resulting shot, with the player who had the chance being the recipient of the ball. Touches of the ball by opposing players after the setup has been executed are considered to be insignificant, provided the player who had the chance would presumably have gained ball control in a comparable match situation without these touches occurring.

		An indirect chance setup is a ball action during which the direction of the ball is significantly affected by an opposing player before reaching the player who had the chance, provided the player who had the chance would presumably not have gained ball control in a comparable match situation without these touches occurring. An indirect chance setup is also recorded if the ball hits the goal frame, is blocked by an opposing player or is saved by the goalkeeper before reaching the player who had the chance. If, in this case, the player who had the chance and the player that played the preceding ball are one and the same, the indirect chance setup is awarded to the player who performed the last ball action before the player who had the chance (usually the player that played the ball to the player who had the chance). Other indicates that the ball was played directly to the player who had the chance by a teammate, whose primary intention was not to set up a chance (e.g. a pass to the player who had the chance in the team's own half, after which the player who had the chance goes on an individual run with the ball before creating a chance without a resulting shot; a player is hit by a shot without becoming active). A free kick chance setup indicates that the player who had the chance receives the ball from a free kick layoff at a free kick. If the chance without a resulting shot occurs directly after a penalty kick or a free kick executed as a shot, the following rules apply: If the player who had the chance is the player that took the penalty kick or free kick, he is awarded a fouled player chance setup, provided no touch of the ball without ball control was performed by any of the player's teammates before the chance (e.g. if a teammate in the wall is hit by the initial shot). If the player who had the chance is not the player that took the penalty kick or free kick, then the player that took the initial shot from the set piece is awarded an indirect chance setup, provided no touch of the ball without ball control was performed by any of the player's teammates before the chance.
Fouled player chance setup	a) reference person ID b) no entry	<i>Fouled player chance setup</i> indicates, when a chance without a resulting shot occurs after a penalty kick or free kick, which was awarded due to a foul against an opposing player. This applies to free kicks that are directly executed as shots or when a free kick layoff leads to a shot. In these cases, the attribute fouled player chance setup refers to the player that was fouled to win the set piece in question. If the manifestation no entry has been used for the attribute chance setup player, then the fouled player can be deemed to have set up the chance (depending on the understanding of what constitutes a setup). This is also the case if the manifestation free kick has been used for the attribute chance setup type.
Ball won		<i>Ball won</i> is a ball action during which a player of the team without possession wins the ball from an opponent through an active tactical action, after which he has certain ball control. In addition to the match event ball won, a match event ball played, shot, own goal or other ball action is to be recorded for the player that won the ball.

Team	a) reference club ID	<i>Team</i> refers to the team of the player that won the ball.
Player	a) reference person ID b) no entry	<i>Player</i> refers to the person that has certain ball control after winning the ball.
Phase of possession number	a) numeric b) no entry	<i>Phase of possession number</i> returns the phase of possession in question.
Type	a) interception by outfield player b) shot blocked with subsequent ball control c) ball saved by goalkeeper d) opponent dispossessed	<i>Type</i> refers to the manner in which the ball was won, from the point of view of the player that won the ball. A distinction is made between the following cases: Interception by outfield player indicates that an outfield player won the ball after an opponent's ball played. Ball saved by goalkeeper indicates that a goalkeeper won the ball after an opponent's ball played or shot. Shot blocked with subsequent ball control indicates that a player blocked an opponent's shot before gaining certain ball control. Opponent dispossessed indicates that a player performed a challenge of the opponent dispossessed type before or whilst winning of the ball and that he is in certain ball control after the challenge.
Solo run before a shot		<i>Solo run before a shot</i> indicates that a player dribbled with the ball for more than 20 metres in the direction of the opponents' goal, regardless of whether the player in question chose the shortest route or rounded opposing players during his action. A solo run before a shot is to be recorded by the match data collector if the solo run is the match situation preceding a shot or own goal. The player that performs the solo run before a shot is assigned an additional ball action. The decision on which ball action to assign should be based on the action that took place.
Team	a) reference club ID	<i>Team</i> refers to the team of the player that performed the solo run.
Player	a) reference person ID	<i>Player</i> refers to the person that performed the solo run.
Fair play		<i>Fair play</i> refers to a ball action, with which a player deliberately relinquishes his team's certain ball control.
Team	a) reference club ID	<i>Team</i> refers to the team of the player that performed the fair play action.
Player	a) reference person ID	<i>Player</i> refers to the person that performed the fair play action.
Phase of possession number	a) numeric	<i>Phase of possession number</i> returns the phase of possession in question.

Other ball action		<i>Other ball action</i> is a player action which cannot be classified as one of the following ball actions: ball played, shot, own goal, own goal prevented, fair play, ball won or challenge during which a one-on-one is successfully performed. For example, other ball action applies to the following situations: A player in certain ball control slips or trips over the ball without an opponent's involvement. A player receives the ball from a teammate and loses the ball before he can play it. The game is halted or the final whistle is blown whilst the player has ball control. A player touches the ball, e.g. when he is hit by a shot. A player performs a clearance.
Team	a) reference club ID	<i>Team</i> refers to the team of the player that performed the other ball action.
Player	a) reference person ID	<i>Player</i> refers to the team of the player that performed the other ball action.
Clearance	a) yes b) no entry	<i>Clearance</i> is a ball action during which a player manages to prevent or prevent a scoring threat to his own team's goal, accepting that his team will lose ball control. It is essential that his ball action cannot be classified as a ball played and that he consciously reacts to the ball. If a player prevents a corner kick from being awarded by clearing the ball into touch or to a teammate before it crosses the goal line, then this action is classified as a clearance.
Phase of possession number	a) numeric	<i>Phase of possession number</i> returns the phase of possession in question.
Offside		<i>Offside</i> is a player action after which the referee gives an offside decision in accordance with the DFB football rules. If several players are offside at the same time, the player is to be recorded that is positioned closest to the place where the subsequent free kick is executed, according to the match data collector.
Team	a) reference club ID	<i>Team</i> refers to the team of the player that caused the offside decision.
Player	a) reference person ID	<i>Player</i> refers to the person that caused the offside decision. The player that was caught offside is to be recorded, not the player that played the ball.
Challenge		<i>Challenge</i> is a player action during which two players of opposing teams are physically capable of either gaining or receiving ball control and attempt to do so. A challenge requires one of the two players to touch the ball or to foul the opposing player. Every challenge has a loser and a winner. The winner is the player that determines the direction of the ball at the end of the challenge. If the match event challenge is ended by the referee's decision to award a foul, which is classified as the foul type opponent fouled, then the player that was fouled is deemed to have won the challenge. If several players are involved in a challenge, then the challenge is only assigned to two of those players. The two players chosen are the player that determines the direction of the ball at the end of the challenge (winner) and the player that is positioned closest to the winner (loser). In addition to the event challenge, a match event ball played,

		shot, own goal, ball won or other ball action is to be recorded for the player or players that were involved in and played the ball during the challenge.
Winner team	a) reference club ID	<i>Winner team</i> refers to the team of the player that won the challenge.
Winner	a) reference person ID	<i>Winner</i> refers to the player that won the challenge.
Winner role	a) in ball control b) not in ball control	<i>Winner role</i> indicates whether the winner engaged in a challenge having been in ball control or having not been in ball control. For every challenge there is a player who (or whose team) had certain ball control and a player who (or whose team) did not have ball control prior to the challenge. The team that last had certain ball control before the challenge is “in ball control”, the other team is “not in ball control”.
Loser team	a) reference club ID	<i>Loser team</i> refers to the team of the player that lost the challenge.
Loser	a) reference person ID	<i>Loser</i> refers to the player that lost the challenge.
Loser role	a) in ball control b) not in ball control	<i>Loser role</i> indicates whether the loser engaged in a challenge having been in ball control or having not been in ball control.
Type	a) opponent rounded b) ball action carried out c) fouled d) opponent dispossessed e) challenge during release of the ball f) possession retained during challenge	<i>Type</i> refers to the type of challenge. A distinction is made between the following cases: • Opponent rounded: a player in ball control stays in ball control after the challenge, having left the opposing player behind him. Situations where the opponent is not able to gain possession (e.g. when the ball is “flicked” over the opponent) are also to be recorded as challenges. • Ball action carried out: applies when none of the players involved in the challenge are in ball control at the start of the challenge (e.g. aerial challenges, challenges for the first touch of the ball) and one player determines the direction of the ball at the end of the challenge. • Fouled: the referee called a foul. • Opponent dispossessed: a player not in ball control dispossesses the opposing player in ball control. • Challenge during release of the ball applies when shots or balls played are forced or blocked during challenges. A challenge is only recorded, if the ball played or shot travels through the area that the defending player is attempting to cover from a tactical perspective. All other cases are not recorded as challenges. • Possession retained during challenge: applies when one of the players involved in the challenge has certain ball control at the start of the challenge and manages to retain it, despite the efforts to dispossess him of the opponent involved in the challenge.

		If several challenge types apply to the same challenge, only one challenge type is to be recorded. This is only possible if none of the players involved in the challenge were in certain ball control before the challenge. In this case, the decision on which challenge type to record should occur in the following order: fouled, opponent rounded, challenge during release of the ball, ball action carried out, opponent dispossessed, ball retained during challenge.
Challenge subcategory	a) aerial b) ground	<i>Challenge subcategory</i> indicates whether the challenge took place at ground level or in the air. Aerial is used when the ball is positioned at head level or higher and if at least one of the players involved in the challenge is not touching the ground during the challenge. Furthermore, none of the players involved in an aerial challenge may have ball control at the start of the challenge. Individual cases may be deemed to be aerial challenges by the match data collector due to their tactical nature, despite both (all) players touching the ground during the challenge (e.g. players running backwards after misjudging the flight of the ball, which is still positioned at head level).
One-on-one	a) successful b) not successful c) no entry	<i>One-on-one</i> indicates whether a player in certain ball control attempts to round an opposing player during the challenge. For all other challenges the manifestation no entry is used. Successful or not successful indicates whether the player in certain ball control managed to round the opponent or not. If a player is fouled during a one-on-one, it is classified as successful.
One-on-one side	a) on the left b) on the right c) other c) no entry	<i>One-on-one side</i> refers to the side that the player in certain ball control wants to pass or passes to get in behind the opposing player involved in the challenge, from the point of view of the player in ball control. If the side cannot be determined, the manifestation other is to be used. The attribute one-on-one side must be assigned for all challenges of the “opponent rounded” type. The attribute must also be assigned for challenges of the “opponent dispossessed” and “fouled” types, provided the manifestation no entry has not been used for the attribute one-on-one in these cases.
One-on-one type	a) close control b) knocked past c) no entry	<i>Close control</i> indicates that the player that performed the one-on-one had multiple touches of the ball during the one-on-one - or that it was his intention to do so – and that he retained certain ball control during the challenge. <i>Knocked past</i> indicates that the player that performed the one-on-one gave up certain ball control in order to regain ball control after getting in behind the opposing player. The attribute <i>one-on-one type</i> is used if “no entry” was recorded for the attribute <i>one-on-one</i> .
Change of possession	a) yes b) no	<i>Change of possession</i> indicates the subsequent course of the ball after the challenge. Yes indicates that the player not in ball control or one of his teammates gains certain ball control or could have done so easily immediately after the challenge. The manifestation no should be used in all other cases. In the special case of a player deliberately relinquishing certain ball control, e.g. by taking a shot, and a challenge occurring during this action, the manifestation no is to be used. This is due to the fact that the player did not intend to retain ball control for his team during this action.
Winner action	a) goal scored b) assist provided c) shot taken d) shot set up e) no entry	<i>Winner action</i> refers to the action executed by the winner of the challenge immediately after the challenge. Goal scored indicates that the winner of the challenge scored a goal immediately after the challenge. Assist provided indicates that the winner of the challenge provided an assist immediately after the challenge, regardless of the type of assist or own goal assist.

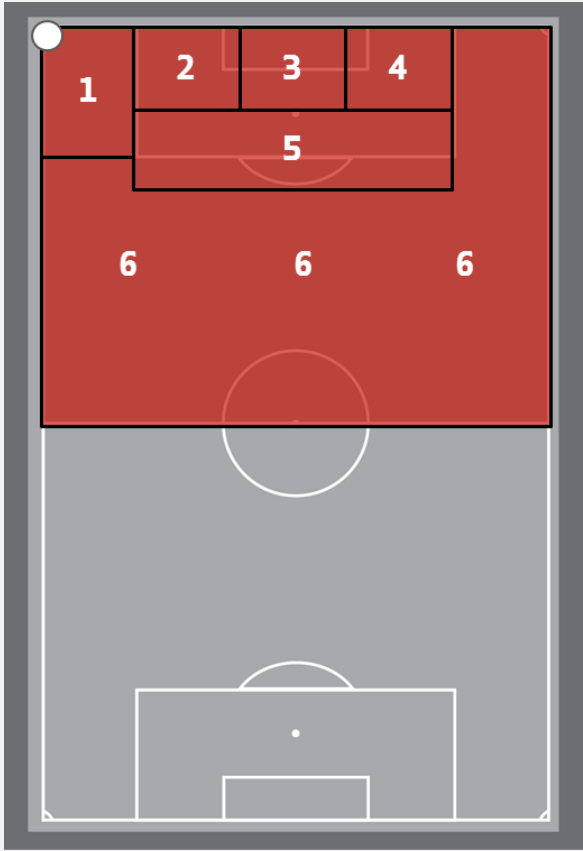
		Shot taken indicates that the winner of the challenge took a shot immediately after the challenge (which did not result in a goal). Shot set up indicates that the winner of the challenge set up a shot immediately after the challenge, regardless of the type of shot setup.
Phase of possession number	a) numeric b) no entry	If the situation in question is a challenge for which the attribute one-on-one has been classified as successful, then the attribute <i>phase of possession number</i> must also be assigned accordingly. In all other situations the manifestation no entry is used.
Goalkeeper involvement	a) yes b) no	<i>Goalkeeper involvement</i> indicates whether one of the players involved in the challenge is a goalkeeper who is permitted to use his hands in the situation in question. Situations involving a goalkeeper and an outfield player are only classified as challenges if both have equal opportunity to win the challenge. The match data collector is responsible for assessing whether equal opportunity is given. A situation is not recorded as a challenge if the opposing outfield player pulls out of the challenge (e.g. because he realises that the goalkeeper would safely catch the ball).
Foul		<i>Foul</i> refers to a match event after which the referee decides that a foul or misconduct has occurred.
Fouling team	a) reference club ID	<i>Fouling team</i> refers to the team of the player that committed the foul.
Fouling player	a) reference person ID	<i>Fouling player</i> refers to the person that committed the foul.
Fouled team	a) reference club ID b) no entry	<i>Fouled team</i> refers to the team of the player that was fouled.
Fouled player	a) reference person ID b) no entry	<i>Fouled player</i> refers to the person that was fouled.
Foul type	a) opponent fouled b) handball c) other rule violation	<i>Foul type</i> refers to the type of foul being penalised. If no player is fouled during a rule violation, then the manifestation no entry is used for the attributes fouled team and fouled player. The foul type “handball” or “other rule violation” is used for these situations. If a player is fouled, then the foul type “opponent fouled” is used.
Fouling player action	a) sliding tackle b) holding c) elbow d) no entry	<i>Fouling player action</i> refers to the action of the player who is being penalised for a foul. Sliding tackle indicates whether the fouling player committed the foul through a slide tackle. Holding indicates whether the fouling player committed the foul by holding an opponent or pulling an opponent’s jersey. Elbow indicates whether the fouling player committed the foul by elbowing an opponent. The manifestation no entry is used for all other actions.

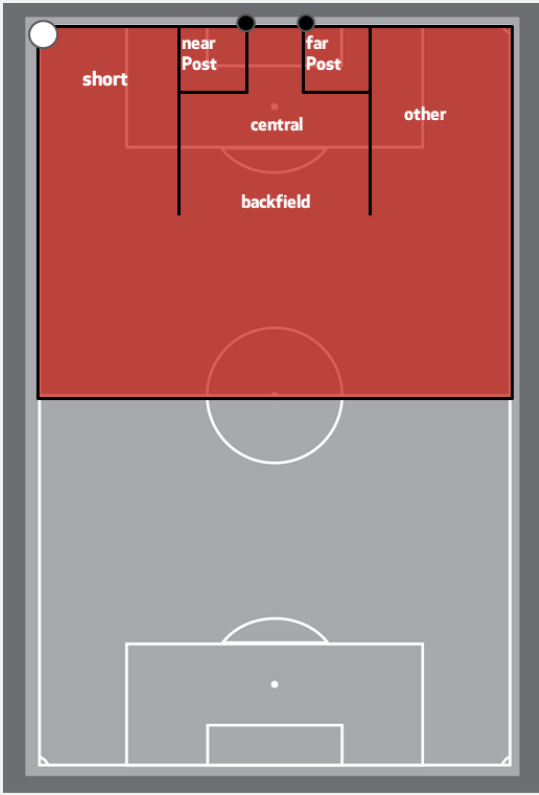
Nutmeg		<i>Nutmeg</i> refers to a situation during which the ball is played through the legs of an outfield player from short distance (e.g. during a challenge or with a pass). Only clear and obvious nutmegs are recorded, as the majority of cases should be deliberate actions. For example, a player that has the ball played through his legs during a sliding tackle with wide-spread legs is usually not deemed to have been nutmegged (only if the player's feet are still touching the ground and the player that executed the nutmeg clearly intended to do so). Any ball that is played through a goalkeeper's legs is recorded as a nutmeg. The player that executes the nutmeg is assigned an additional ball action. The decision on which ball action to assign should be based on the action that took place.
Executing team	a) reference club ID	<i>Executing team</i> refers to the team of the player that executed the nutmeg.
Affected team	a) reference club ID	<i>Affected team</i> refers to the team of the player was nutmegged.
Executing player	a) reference person ID	<i>Executing player</i> refers to the person that executed the nutmeg.
Affected player	a) reference person ID	<i>Affected player</i> refers to the person that was nutmegged.
Spectacular action		<i>Spectacular action</i> indicates whether the ball was played using an overhead kick or a backheel. The player that executes the spectacular action is assigned an additional ball action. The decision on which action to assign should be based on the respective action.
Team	a) reference club ID	<i>Team</i> refers to the team of the player that executed the spectacular action.
Player	a) reference person ID	<i>Player</i> refers to the person that executed the spectacular action.
Type of action	a) overhead kick/sideways scissor kick b) backheel	<i>Type of action</i> indicates whether the ball was played using an "overhead kick/sideways scissor kick" or a "backheel". When the ball is played using a back heel (with the heel of the foot/boot) the ball must actually be played or laid off (for example, if a player uses the heel or sole of his foot/boot during a one-on-one situation this is not classified as a backheel).
Other player action		<i>Other player action</i> refers to a situation that is not classified as one of the following actions: ball played, shot, own goal, own goal prevented, other ball action, fair play, challenge, ball won, offside, foul, nutmeg, chance without a resulting shot, solo run before a shot or spectacular action.
Team	a) reference club ID	<i>Team</i> refers to the team of the player that executes the other player action.

Player	a) reference person ID	<i>Player</i> refers to the person that executes the other player action.
Change of captain's armband	a) yes b) no	<i>Change of captain's armband</i> indicates whether the other player action in question is a change of captain's armband during the match. The attribute player should refer to the player that receives the captain's armband during the change.
Maximum number of substitutions used	a) yes b) no	<i>Maximum number of substitutions used</i> indicates whether the other player action in question is a player leaving the pitch without being replaced by a substitute, due to the maximum number of substitutions having already been used.
Goes in goal	a) yes b) no	<i>Goes in goal</i> indicates whether an outfield player takes up the goalkeeper's position during the match. The attribute player should refer to the player that takes over as goalkeeper in this situation.
Deflection		Refers to the ball being deflected by something other than a player involved in the game or the pitch itself during the match.
Type	a) left post b) right post c) crossbar d) corner flag e) referee f) other	Returns the cause of the <i>deflection</i> from the point of view of the team in ball control. The manifestation corner flag is not further distinguished. "Other" would refer to the ball being deflected by a camera positioned above the pitch or by a video cube, for example.
Kick-off		<i>Kick-off</i> refers to the match event which may be executed on the referee's whistle, in accordance with the DFB football rules, after the match events successful shot or own goal have occurred. The kick-off also represents the start of the respective periods of play.
Period of play	a) first half b) second half c) first half of extra time d) second half of extra time e) no entry	<i>Period of play</i> indicates whether the kick-off takes place at the start of the first/second half or at the start of the first/second half of extra time, respectively. The manifestation no entry is used for kick-offs after goals.
Execution type	a) ball played b) shot c) fair play	<i>Execution type</i> indicates whether the kick-off was executed as a ball played, as fair play or as a shot.
Team left	a) reference club ID	<i>Team left</i> refers to the team in the left half of the pitch at kick-off.

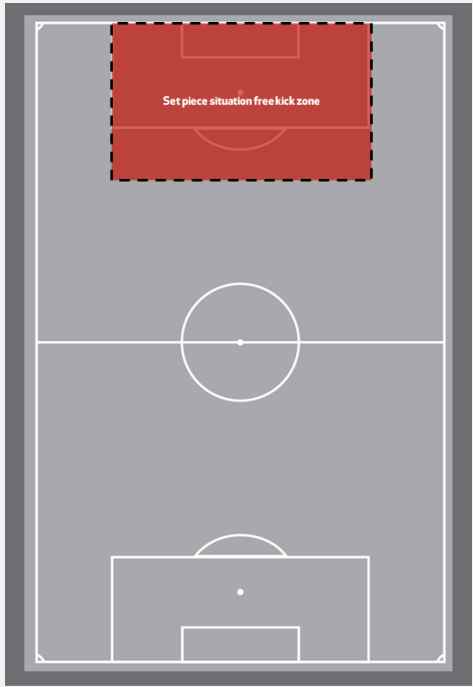
Team right	a) reference club ID	<i>Team right</i> refers to the team in the right half of the pitch at kick-off.
Goal kick		<i>Goal kick</i> refers to a situation where the referee awards a goal kick in accordance with the DFB football rules and a goal kick is then executed.
Execution type	a) ball played b) shot c) fair play	<i>Execution type</i> indicates whether the goal kick was executed as a ball played, as fair play or as a shot.
Team	a) reference club ID	<i>Team</i> refers to the team that executed the goal kick.
Decision time	a) time stamp	<i>Decision time</i> returns the time when the referee awarded the goal kick or when the decision resulted from the course of the match.
Throw-in		<i>Throw-in</i> refers to a situation where the referee awards a throw-in in accordance with the DFB football rules and a throw-in is then executed.
Execution type	a) ball played b) fair play c) no entry	<i>Execution type</i> indicates whether the throw-in was executed as a ball played or as fair play. If the execution type of the throw-in is a ball played classified as a cross, the term throw-in cross is used.
Team	a) reference club ID	<i>Team</i> refers to the team that executed the throw-in.
Execution side	a) left b) right	<i>Execution side</i> indicates whether the ball was thrown in from the left or the right side of the pitch, from the point of view of the executing team's direction of play.
Foul throw	a) yes b) no entry	<i>Foul throw</i> indicates whether the referee has awarded a foul throw, in accordance with the DFB football rules.
Foul throw team	a) reference club ID b) no entry	<i>Foul throw team</i> refers to the team that executed the foul throw.
Foul throw player	a) reference person ID b) no entry	<i>Foul throw player</i> refers to the person that executed the foul throw.
Phase of possession number	a) numeric b) no entry	The attributes foul throw, foul throw team and foul throw player are only to be recorded if the referee awards a foul throw in accordance with the DFB football rules. In this case, the manifestation no entry is used for the attribute execution type, whilst the attribute <i>phase of possession number</i> is recorded accordingly.

Decision time	a) time stamp	<i>Decision time</i> returns the time when the referee awarded the throw-in or when the decision resulted from the course of the match.
Corner kick		<i>Corner kick</i> refers to a situation where the referee awards a corner kick in accordance with the DFB football rules and a corner kick is then executed.
Execution type	a) ball played b) shot c) fair play	<i>Execution type</i> indicates whether the goal kick was executed as a ball played, as a shot or as fair play.
Team	a) reference club ID	<i>Team</i> refers to the team that executed the corner kick.
Execution side	a) left b) right	<i>Execution side</i> indicates whether the corner kick was taken from the left or the right side of the pitch, from the point of view of the executing team's direction of play.
Placement	a) opponent hit b) short c) near post d) central e) far post f) edge of the box g) other h) out of play	<i>Placement</i> refers to the direction of the corner kick. Opponent hit indicates that an opposing player touched the ball in zone 1 immediately after the corner kick was taken. Short (1), near post (2), central (3), far post (4), edge of the box (5) and other (6) refer to the position of the first ball action after the corner kick has been taken. This does not apply if the ball is played straight out of play, without another player touching the ball. In this case the manifestation out of play is to be used. The zones are defined as follows: 1 to the near side of the penalty area 2 from the near side of the penalty area to the near post 3 between the goalposts, from the goal line to the penalty spot 4 from the far post to the far side of the penalty area 5 between the sides of the penalty area, from the penalty spot to the furthest point of the penalty arc 6 from the sides of the penalty area to the touch line and the remaining space to the half-way line

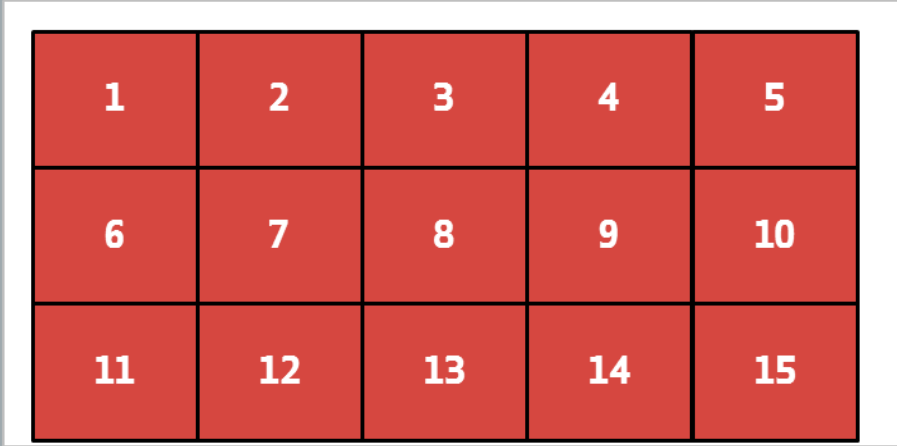
		 Figure 9: Overview of placement zones at corner kicks
Target area	a) opponent hit b) short c) near post d) central e) far post f) backfield g) other	<i>Target area</i> refers to the intended direction of the corner kick, which may differ from the placement, and is identical to the attribute placement regarding its manifestations and respective descriptions. Regardless of the executing player's intention, the manifestations opponent hit and out of play are recorded in the same way for the attribute target area as for the attribute placement, as the executing player's intention is not always apparent in these cases.

	h) out of play	 Figure 10: Overview of placement zones at corner kicks
Rotation	a) inswinging b) outswinging c) other	<i>Rotation</i> indicates whether the ball's trajectory is being significantly altered by its lateral rotation - either towards goal (inswinging) or away from goal (outswinging) - or whether the ball is played in any other way.
Cover at the post	a) near post b) far post c) both posts d) no cover	<i>Cover at the post</i> indicates whether the team defending the corner has outfield players covering the near post, far post or both posts.

Decision time	a) time stamp	<i>Decision time</i> returns the time when the referee awarded the corner kick or when the decision resulted from the course of the match.
Start frame	a) frame number	Indicates the start of the set piece situation by returning the frame number provided by the official positional data collection. Start of the set piece situation means that it is apparent that at least one player of the team in ball control, who is not the person taking the corner kick, significantly changes his position significantly in direct relation to and in anticipation of the corner kick, or at the latest when the corner kick is taken.
End frame	a) frame number	Indicates the end of the set piece situation by returning the frame number provided by the official positional data collection. End of the set piece situation means that a match situation occurs after the corner kick, which is no longer directly connected to the corner kick. The match data collector is responsible for assessing whether this is the case. If the game is halted, the set piece situation is deemed to have ended.
Free kick		<i>Free kick</i> refers to a situation where the referee awards a free kick in accordance with the DFB football rules and a free kick is then executed.
Team	a) reference club ID	<i>Team</i> refers to the team that executed the free kick.
Execution type	a) ball played b) shot c) fair play	<i>Execution type</i> indicates whether the free kick was executed as a ball played, as a shot or as fair play.
Execution specification	a) direct b) indirect	<i>Execution specification</i> indicates whether the referee has awarded a direct or an indirect free kick in accordance with the DFB football rules.
Decision time	a) time stamp	<i>Decision time</i> returns the time when the referee awarded the free kick or when the decision resulted from the course of the match.
Start frame	a) frame number b) no entry	Indicates the start of the set piece situation by returning the frame number provided by the official positional data collection. The start frame is recorded when the free kick is either executed as a shot (including layoffs) or is executed from within, played into or through the area depicted (figure 10a). The attacking quarter, or half the distance between the goal line and the half-way line, represents the rear boundary of the area depicted, whilst the sides of the penalty area represent the side boundaries. Start of the set piece situation means that it is apparent that at least one player of the team in ball control, who is not the person taking the free kick, significantly changes his position significantly in direct relation to and in anticipation of the free kick, or at the latest when the free kick is taken.

		 Figure 10a: Set piece situation free kick zone
End frame	a) frame number b) no entry	Indicates the end of the set piece situation by returning the frame number provided by the official positional data collection. The end frame is recorded if a start frame has been recorded. End of the set piece situation means that a match situation occurs after the free kick, which is no longer directly connected to the free kick. The match data collector is responsible for assessing whether this is the case. If the game is halted, the set piece situation is deemed to have ended.
Penalty kick		<i>Penalty kick</i> refers to a situation where the referee awards a penalty kick in accordance with the DFB football rules.
Team	a) reference club ID	<i>Team</i> refers to the team that executed the penalty kick.

Prospective penalty taker	a) reference person ID	<i>Prospective penalty taker</i> refers to the player that is expected to take the awarded penalty. This is also the case for a penalty during a penalty shootout.
Execution type	a) ball played b) shot c) fair play	<i>Execution type</i> indicates whether the penalty kick was executed as a ball played, as a shot or as fair play.
Fouled player	a) reference person ID b) no entry	<i>Fouled player</i> refers to the player that is fouled immediately before the penalty kick is awarded, provided this foul is the reason the penalty kick was awarded. If the referee awards a penalty kick for other reasons, then the manifestation no entry is used.
Player responsible	a) reference person ID b) no entry	<i>Player responsible</i> refers to the player who performed the action that led the referee to award a penalty kick. The manifestation no entry is used for penalty kicks during a penalty shootout.
Goalkeeper movement	a) left b) right c) none	<i>Goalkeeper movement</i> indicates the goalkeeper's plan of action before or during the penalty taker's touch of the ball. The manifestations left and right, from the goalkeeper's point of view, indicate whether the goalkeeper made an irreversible movement to cover the respective goal zones (right = zones 1, 2, 6, 7, 11, 12; left = zones 4, 5, 9, 10, 14, 15). It is essential that covering the other zones, respectively, is no longer possible after the goalkeeper's movement. The manifestation none is to be used if the goalkeeper does not perform an irreversible movement.

		 Figure 8: Overview of goal zones from the point of view of the attacking team
Goalkeeper behaviour	a) no rule violation b) rule violation	<i>Goalkeeper behaviour</i> indicates whether the goalkeeper of the defending team committed a rule violation at the penalty kick, in accordance with the DFB football rules.
Encroachment	a) teammate b) opposing player c) both teams' players c) no player	<i>Encroachment</i> refers to players entering the penalty area or the penalty arc before the execution of the penalty kick. Teammate indicates that at least one of the penalty taker's teammates encroached. Opposing player indicates that at least one of the opposing players encroached, from the point of view of the penalty taker. Both teams' players indicates that at least one player of either team encroached. No player indicates that no player encroached.
Retaken penalty kick	a) yes b) no entry	<i>Retaken penalty kick</i> indicates whether the penalty kick was retaken.

Retaken penalty kick player	a) reference player ID b) no entry	Retaken penalty kick player refers to the person that executed the retaken penalty kick.																																																																																																																
Retaken penalty kick target zone	a) numeric b) no entry	Retaken penalty kick target zone returns the target zone at penalty kick of the retaken penalty kick. Target zone at penalty kick indicates the zone which the ball would pass through once it reaches the goal line, if it was not deflected by the opposing goalkeeper. <table><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td></tr><tr><td>15</td><td>16</td><td>17</td><td>18</td><td>19</td><td>20</td><td>21</td><td>22</td><td>23</td><td>24</td><td>25</td><td>26</td><td>27</td><td>28</td></tr><tr><td>29</td><td>30</td><td>31</td><td>32</td><td>33</td><td>34</td><td>35</td><td>36</td><td>37</td><td>38</td><td>39</td><td>40</td><td>41</td><td>42</td></tr><tr><td>43</td><td>44</td><td>45</td><td>46</td><td>47</td><td>48</td><td>49</td><td>50</td><td>51</td><td>52</td><td>53</td><td>54</td><td>55</td><td>56</td></tr><tr><td>57</td><td>58</td><td>59</td><td>60</td><td>61</td><td>62</td><td>63</td><td>64</td><td>65</td><td>66</td><td>67</td><td>68</td><td>69</td><td>70</td></tr><tr><td>71</td><td>72</td><td>73</td><td>74</td><td>75</td><td>76</td><td>77</td><td>78</td><td>79</td><td>80</td><td>81</td><td>82</td><td>83</td><td>84</td></tr><tr><td>85</td><td>86</td><td>87</td><td>88</td><td>89</td><td>90</td><td>91</td><td>92</td><td>93</td><td>94</td><td>95</td><td>96</td><td>97</td><td>98</td></tr><tr><td>99</td><td>100</td><td>101</td><td>102</td><td>103</td><td>104</td><td>105</td><td>106</td><td>107</td><td>108</td><td>109</td><td>110</td><td>111</td><td>112</td></tr></table> Figure 7: Overview of target zones at a penalty kick from the point of view of the attacking team	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112
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Retaken penalty kick result	a) scored b) not scored c) no entry	Retaken penalty kick result indicates whether the retaken penalty kick resulted in a goal.																																																																																																																
Retaken penalty kick reason	a) executing player's behaviour b) goalkeeper's behaviour c) other player's behaviour	Retaken penalty kick reason indicates why the referee decided to have the penalty kick retaken.																																																																																																																

	d) other e) no entry	
Decision time	a) time stamp b) no entry	<i>Decision time</i> returns the time when the referee awarded the penalty kick or when the decision resulted from the course of the match. The manifestation no entry is used for penalty kicks during a penalty shootout.
Start of penalty shootout		<i>Start of penalty shootout</i> returns the time when the penalty shootout started, indicated by the referee blowing his whistle or giving another sign.
Whistle ending the period of play		<i>Whistle ending the period of play</i> refers to the referee ending the respective period of play by blowing his whistle.
Period of play	a) first half b) second half c) first half of extra time d) second half of extra time e) penalty shootout	<i>Period of play</i> indicates which period of play has been ended (e.g. first half, second half, first half of extra time, second half of extra time, penalty shootout).
Period of play result	a) alphanumeric	<i>Period of play result</i> returns the result of the respective period being ended.
Abandonment	a) yes b) no	<i>Abandonment</i> indicates whether the referee abandoned the game in accordance with the DFB football rules.
Reason	a) text entry	<i>Reason</i> refers to the reason for the abandonment of the match.
Substitution		<i>Substitution</i> refers to the replacement of a player by a substitute. The execution of the substitution is determined by the DFB football rules. The substitution should be recorded at the actual time of substitution, not at the time the substitution is announced.
Team	a) reference club ID	<i>Team</i> refers to the team making the substitution.
Player brought off	a) reference person ID	<i>Player brought off</i> refers to the player who is being replaced.
Substitute brought on	a) reference person ID	<i>Substitute brought on</i> refers to the player who is coming on as a substitute.

Player position	a) GK: goalkeeper b) LB: left back c) LCB: left centre back d) CB: centre back e) RCB: right centre back f) RB: right back g) WLDM: wide-left defensive midfielder h) LDM: left defensive midfielder i) CDM: central defensive midfielder j) RDM: right defensive midfielder k) WRDM: wide-right defensive midfielder l) LM: left midfielder m) HLM: half-left midfielder n) CM: central midfielder o) HRM: half-right midfielder p) RM: right midfielder q) LAM: left attacking midfielder r) HLAM: half-left attacking midfielder s) CAM: central attacking midfielder t) HRAM: half-right attacking midfielder u) RAM: right attacking midfielder v) SS: supporting striker w) LW: left wide player x) LS: left striker y) CF: centre forward z) RS: right striker - aa) RW: right wide player	<i>Player position</i> is the position assigned to the player being brought on as a substitute.
Referee substitution		<i>Referee substitution</i> is a match event during which the referee is substituted in accordance with the DFB football rules. The referee substitution should be recorded at the actual time of substitution, not at the time the substitution is announced.

Referee brought off	a) reference person ID	<i>Referee brought off</i> refers to the referee who is being replaced.
Referee brought on	a) reference person ID	<i>Referee brought on</i> refers to the referee who is coming on as a substitute.
Caution or dismissal		<i>Caution or dismissal</i> refers to a match event during which the referee decides to caution or dismiss a player in accordance with the DFB football rules. The caution or dismissal should be recorded at the actual time of cautioning or dismissal (meaning the showing of the card), not at the time when the player action in question occurred.
Team	a) reference club ID	<i>Team</i> refers to the team of the player that has been cautioned or dismissed by the referee in accordance with the DFB football rules.
Player	a) reference person ID	<i>Player</i> refers to the person that has been cautioned or dismissed by the referee in accordance with the DFB football rules.
Card	a) yellow card b) second yellow card in the same match c) straight red card d) yellow card during penalty shootout	<i>Card</i> indicates whether the player that has been cautioned or dismissed received a yellow card, a second yellow card in the same match or a straight red card.
Card evaluation	a) yellow card b) second yellow card in the same match c) straight red card d) annulled e) yellow card not counted	<i>Card evaluation</i> refers to the manner in which the caution or dismissal that the referee decided on is permanently recorded. “Annulled” indicates that a caution or dismissal was overturned after the final whistle (e.g. through a sports court ruling). “Yellow card not counted” indicates that a yellow card will not count towards a player’s total number of bookings, as the player in question was subsequently sent off. If a caution or dismissal is both annulled and not counted, the manifestation annulled is to be used.
Reason	a) foul b) unsporting behaviour c) violent conduct d) handball e) other	<i>Reason</i> refers to the reason for the referee’s decision to caution or dismiss a player, which has been entered in the official match report.
Caution or dismissal of team officials		<i>Caution or dismissal of team officials</i> refers to a match event during which the referee cautions or dismisses a team official in accordance with the DFB football rules.

Team	a) reference club ID	<i>Team</i> refers to the team of the team official that has been cautioned or dismissed by the referee.
Team official cautioned or dismissed	a) reference person ID b) no entry	<i>Team official cautioned or dismissed</i> refers to the team official that has been cautioned or dismissed by the referee.
Card	a) yellow card b) second yellow card in the same match c) straight red card	<i>Card</i> indicates whether the team official that has been cautioned or dismissed received a yellow card, a second yellow card in the same match or a straight red card.
Dropped ball		<i>Dropped ball</i> refers to a match event during which the referee decides to award a dropped ball in accordance with the DFB football rules.
Goal disallowed		<i>Goal disallowed</i> indicates that a goal or own goal was disallowed by the referee.
Team (that would have scored)	a) reference club ID	<i>Team (that would have scored)</i> refers to the team in whose favour the disallowed goal or own goal would have been awarded.
Player (that would have scored)	a) reference person ID	<i>Player (that would have scored)</i> refers to the person that scored the disallowed goal or own goal.
Reason	a) offside b) opponent fouled c) handball d) other	<i>Reason</i> refers to the apparent reason for the referee disallowing the goal or own goal.
Other referee action		<i>Other referee action</i> indicates that the referee has to intervene due to an external influence on the game, in accordance with the DFB football rules, and that none of the following events occurred: • Start of penalty shootout • Whistle ending the period of play • Substitution • Referee substitution • Caution or dismissal • Caution or dismissal of team officials • Dropped ball • Goal disallowed

Reason	a) text entry	<i>Reason</i> refers to the reason for the other referee action.
Start of interruption of play	a) alphanumeric b) no entry	<i>Start of interruption of play</i> returns the exact time when the referee stopped the clock. If no interruption occurs, the manifestation no entry is used. In this case, the term “stopping the clock” indicates that once the interruption has ended, the game will continue from the minute of play during which the clock was stopped. Also, the time that elapsed during the interruption will not be recorded as stoppage time. This is the case, for example, if the referee tells both teams’ players to leave the pitch. The clock is not stopped for players’ or referees’ injury breaks.
End of interruption of play	a) alphanumeric b) no entry	<i>End of interruption of play</i> returns the exact time when the referee restarts the clock after an interruption of play has ended. If no interruption occurs, the manifestation no entry is used.
Video assistant referee review		<i>Video assistant referee review</i> refers to a situation during which the video assistant referee markedly influences the referee’s decision or advises him to review the decision himself.
Video assistant referee	a) reference person ID	<i>Video assistant referee</i> refers to the person named as the video assistant referee in the official match report.
Referee	a) reference person ID	<i>Referee</i> refers to the person named as the referee in the official match report.
Assistant Referee 1	a) reference person ID	<i>Assistant referee 1</i> refers to the person named as the assistant referee 1 in the official match report.
Assistant Referee 2	a) reference person ID	<i>Assistant referee 2</i> refers to the person named as the assistant referee 2 in the official match report.
Start of review	a) time stamp	<i>Start of review</i> returns the exact time of the event that caused the video assistant referee review to occur. If the initial starting action itself is not an event, then the time stamp must be recorded independently.
End of review	a) time stamp	<i>End of review</i> returns the exact time when play continues after the video assistant referee review. If play does not continue after the action has occurred, then the time stamp of the half-time whistle or final whistle is recorded.
Goes to the RRA	a) yes b) no	<i>Goes to the RRA</i> (referee review area) indicates whether the referee goes to the RRA before making the decision.
Team being checked	a) reference club ID	<i>Team being checked</i> refers to the team whose event is being checked.
Opposing team	a) reference club ID	<i>Opposing team</i> refers to the opposing team of the team being checked.
Event	a) goal b) penalty kick c) straight red card d) mistaken player identity e) other	<i>Event</i> refers to the match event being checked during the video assistant referee action. The event in question could be a goal, a penalty kick awarded or not awarded, a straight red card awarded or not awarded, or a case of mistaken player identity. In all other cases the manifestation other is used.

Referee's decision	a) upheld b) overturned	<i>Referee's decision</i> indicates whether the referee upholds or overturns his initial decision after the video assistant referee review.
Final decision	a) event awarded b) event not awarded c) other	<i>Final decision</i> indicates whether the event being checked is awarded or not awarded by the referee after having been checked.
Decision evaluation	a) correct b) controversial c) incorrect	<i>Decision evaluation</i> refers to the quality of the decision to uphold or overturn the initial decision regarding the event. "Correct" is used when the decision was correct. "Controversial" is used if the decision could not be clearly evaluated, but there are reasonable doubts whether the decision was correct. "Incorrect" is used when the decision to uphold or overturn the initial decision regarding the event was obviously incorrect.
Clarification	a) text entry	

3.3 STATISTICAL BASE UNITS FROM MATCH EVENTS

In order to generate an official statistical base unit (e.g. passes from open play), the frequency in which the relevant events occur is calculated (see figure 11). Different types of events can also be combined to form a base unit (e.g. the number of goalkeeper's saves is derived from the number of shots saved and the number of own goals prevented). Likewise, events of the same type - depending on the attributes' manifestations - can be cumulated to form different base units (e.g. corner kicks can be classified as crosses from set pieces or passes from set pieces, depending on the respective execution type).

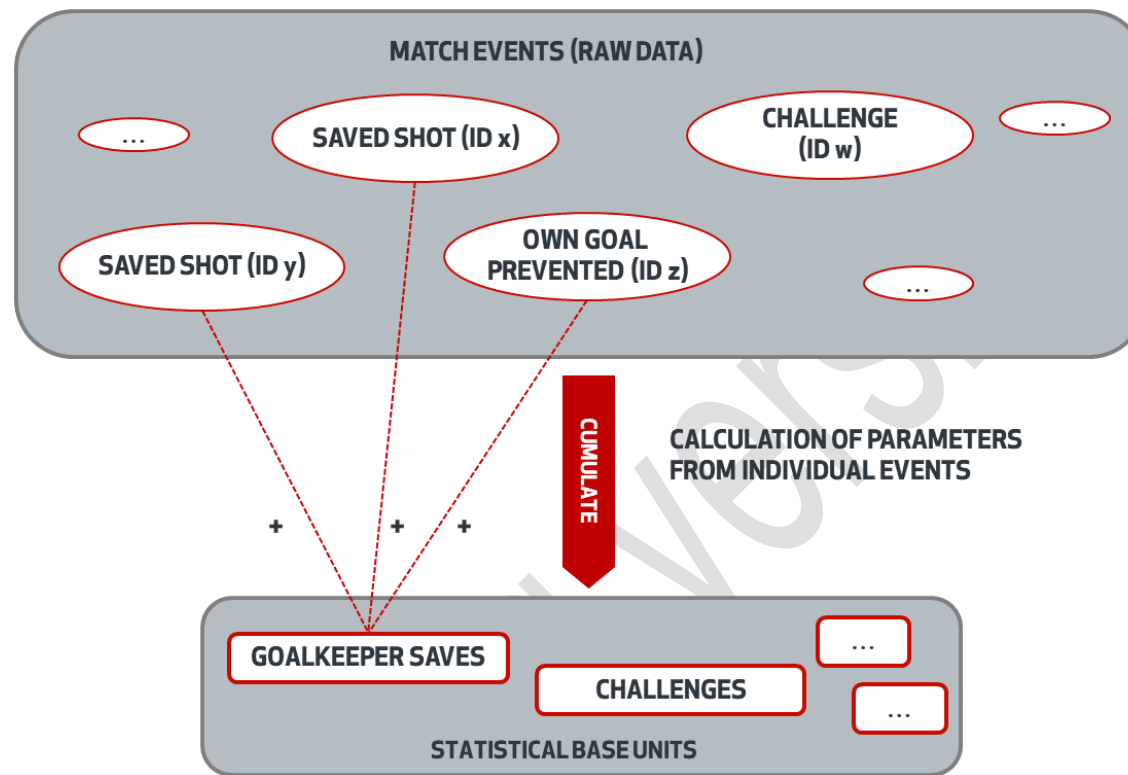


Figure 11: Calculation of statistical base units based on match event raw data (simplified)

Table 1 shows which events and attribute manifestations are added up when calculating a certain base unit (the formalised rule can be found under [1] of the formulary regarding the calculation of statistics from official match data). When using an official statistical base unit, the exact official label defined here should be used. The labels and calculation of base units that have not been listed here have therefore not yet been defined. Enabling uniform use of key statistical variables in sports reporting and other data use processes remains the overall objective. Hence, this catalogue will be continually extended and developed in close cooperation with the relevant stakeholders (especially clubs, the media, and service providers).

Table 1: Statistical base units based on match events
(with the exceptions of cautions and dismissals, match events during a penalty shootout are not considered in the statistical base units)

<i>Official label</i>	<i>Description</i>
Minute of play	The minute of play of a match event is calculated by first subtracting the starting time stamp of the respective period of play from the time stamp of the event in question and then adding the sum of time elapsed in the preceding periods of play without stoppage time (e.g. 0, 45, 90, 105). Additional clarification for the publication of minutes of play: Minutes of play are listed without decimal places. Match events that occur in stoppage time of a period of play are listed in the following format: 45+# or 90+#, 105+# or 120+#.
Phases of possession (quantity)	Represents the total count of phase of possession number attributes that have been assigned a different number.
Possession	[see section statistical base units from positional data]
Ball actions	Represents the total count of all balls played, shots, own goals, balls won, fair play actions, other ball actions as well as challenge during which a one-on-one is successfully performed¹. These challenges are only recorded as a ball action for the player in ball control. Own goals prevented, nutmegs, solo runs before shots and chances without resulting shots are not counted as ball actions². Clarification: ¹ Another ball action (e.g. shot, ball played, ball won) is recorded in all other situations involving a challenge. Otherwise, if this other situation was included, it would be counted twice. ² Another ball action (e.g. ball played, ball won, other ball action) is recorded for an own goal prevented, a nutmeg or solo run before a shot. Otherwise, if own goals, nutmegs and solo runs before shots were included, this action would be counted twice. Chances without resulting shots are either also different ball actions or can even be recorded without a touch of the ball occurring (e.g. if a player misses the ball completely in the attempt to take a shot in front of goal).
Balls played	Represents the total count of all balls played from set pieces and from open play. (For the classification of balls played and passes, refer to figure 12).
Balls played from open play	Represents the total count of all passes from open play, crosses from open play and throw-outs.
Balls played from set pieces	Represents the total count of all passes from set pieces, crosses from set pieces and throw-ins.
Passes from open play	Represents the total count of all passes not played from set pieces, including all kick-outs by the goalkeeper. Throw-outs by the goalkeeper are not included.
Crosses from open play	Represents the total count of all crosses not played from set pieces.

<i>Official label</i>	<i>Description</i>
Balls into the area	Represents the total count of all crosses (without throw-in crosses), low balls into the area and half-field balls into the area. (For the classification of balls into the area, refer to figure 13c).
Balls into the area from open play	Represents the total count of all crosses, low balls into the area and half-field balls into the area not played from set pieces.
Balls into the area from set pieces	Represents the total count of all crosses, low balls into the area and half-field balls into the area played from free kicks and corner kicks.
Throw-in crosses	Represents the total count of all crosses from throw-ins.
Throw-outs	Represents the total count of all throw-outs by the goalkeeper.
Passes from set pieces	Represents the total count of all passes from free kicks, corner kicks, penalty kicks, kick-offs and goal kicks (without passes from throw-ins).
Crosses from set pieces	Represents the total count of all crosses from free kicks and corner kicks (without throw-in crosses)
Passes	Represents the total count of all passes from set pieces and from open play (without passes from throw-ins).
Misplaced passes	Represents the total count of all passes from open play classified as not successful.
Successful passes	Represents the total count of all passes from open play, minus the misplaced passes.
Crosses	Represents the total count of all crosses from set pieces and crosses from open play (without throw-in crosses).
Shots	Represents the total count of all shots (successful shot (goal), shot saved, shot blocked, shot off the goal frame, shot wide and other shot).
Not successful shots	Represents the total count of all shots, minus the successful shots.
Percentage of goals scored	Represents the total count of all successful shots (goals) divided by to the total count of shots.

<i>Official label</i>	<i>Description</i>
Shots on goal	Represents the total count of all successful shots (goals), shots saved, shots off the goal frame and all shots blocked which prevented a goal.
Shots on target	Represents the total count of all shots on goal, minus the shots off the goal frame.
Assists	Represents the total count of all assists of the direct, other and free kick type.
Direct goal involvements	Represents the total count of all successful shots (goals) and direct assists. If a player has also been awarded an indirect goal involvement for the same goal, then only the direct goal involvement is recorded in this case.
Indirect goal involvements	Represents the total count of all contributions to a goal (second assists), all assists of the direct, other and free kick type as well as all fouled player assists. If multiple different assist types listed above have been recorded for the same player and the same goal, then the player is only awarded one indirect goal involvement.
Goal involvements	Represents the total count of all direct and indirect goal involvements. (For the classification of goal involvements, refer to figure 13).
Direct shot involvements	Represents the total count of all shots and direct shot setups. If a player has also been awarded an indirect shot involvement for the same goal, then only the direct shot involvement is recorded in this case.
Indirect shot involvements	Represents the total count of all contributions to a goal (second assists), all shot setups of the direct, other and free kick type as well as all fouled player shot setups. If multiple different shot setup types listed above have been recorded for the same player and the same shot, then the player is only awarded one indirect shot involvement.
Shot involvements	Represents the total count of all direct and indirect shot involvements. (For the classification of shot involvements, refer to figure 13a).
Scoring opportunities	Represents the total count of all involvements in chances and clear-cut chances. (For the classification of scoring opportunities, refer to figure 13b).
Involvement in clear-cut chances	Represents the total count of the following attributes: clear-cut chances with a resulting shot, clear-cut chances without a resulting shot, player responsible for a clear-cut chance and clear-cut chance setups.
Involvement in chances	Represents the total count of all chances without a resulting shot, chances with a resulting shot and chance setups.

<i>Official label</i>	<i>Description</i>
Counterattacks	Represents the total count of all counterattacks before shots (without counterattacks before penalty kicks) and before opponents' own goals.
Goalkeeper's saves	Represents the total count of all shots saved and own goals prevented.
Percentage of shots saved (GK)	Represents the total count of shots saved divided by the sum of successful and shots saved, as well as own goals conceded.
Successful first phase of play (GK)	Represents the total count of all successful and successfully received throw-outs and kick-outs.
Unsuccessful first phase of play (GK)	Represents the total count of all throw-outs and kick-outs, minus the total count of all successful first phases of play.
Challenges	Represents the total count of all defined challenge types (opponent rounded, ball action carried out, fouled, opponent dispossessed, challenge during release of the ball, possession retained during challenge). Challenges involving the goalkeeper are also included.
Percentage of challenges won	Represents the total count of all successful challenges divided by the total count of challenges executed.
Fouls	Represents the total count of all defined foul types of the match event foul, meaning opponent fouled, handball and other rule violation.
Fouls against opponents	Represents the total count of all fouls of the opponent fouled foul type.
Fouls by the opponents	Represents the total count of all fouls of the opponent fouled foul type, from the point of view of the team being fouled.
Handballs	Represents the total count of all fouls of the handball foul type.
Straight red cards/red cards after a second caution in the same match/yellow cards	Represents the total count of all cautions or dismissals of the type in question. In the case of a dismissal by a straight-red card and a dismissal after a second caution in the same match, any caution (yellow card) awarded to that player in the same match is not included. This definition is based on §15 (3) c) of the laws of the game (SpOL).
Set piece situation	Represents the time elapsed (difference) between the start and end frames of the set piece in question.

Match event data contain events which can occur during the match, before the start of the match, in the time between two periods of play or after the match has ended. For example, a caution or dismissal can be awarded before the start of the match, during the half-time break, during the break before extra-time starts or after the match has ended. Substitutions are often made during the half-time break. The following table indicates which periods of play these events are assigned to.

<i>Official label</i>	<i>Description</i>
Substitution	A substitution made in the time between two periods of play should always be assigned to the subsequent period of play.
Caution or dismissal	A caution or dismissal awarded in the time between two periods of play or after the game has ended should always be assigned to the period of play that last ended. A caution or dismissal awarded before the start of the game is assigned to the first half of play.
Change of captain's armband	A change of captain's armband in the time between two periods of play should always be assigned to the subsequent period of play.
Maximum number of substitutions used	If the attribute maximum number of substitutions used is recorded in the time between two periods of play, it should always be assigned to the subsequent period of play.
Referee substitution	A referee substitution in the time between two periods of play should always be assigned to the subsequent period of play.
Caution or dismissal of team officials	Cautions or dismissals of team officials that occur in the time between two periods of play should always be assigned to the subsequent period of play.
Abandonment	An abandonment (of the match) in the time between two periods of play should always be assigned to the period of play that last ended.

BALLS PLAYED					
BALLS PLAYED FROM OPEN PLAY			BALLS PLAYED FROM SET PIECES		
PASSES FROM OPEN PLAY ¹	CROSSES FROM OPEN PLAY	THROW-OUTS	PASSES FROM SET PIECES	CROSSES FROM SET PIECES	THROW-INS

PASSES AND CROSSES							
PASSES			CROSSES				
PASSES FROM OPEN PLAY ^{1 6}	PASSES FROM SET PIECES ⁵		CROSSES FROM OPEN PLAY	CROSSES FROM SET PIECES ⁴			
	FREE KICKS ²	PENALTY KICKS ²	CORNER KICKS ²	GOAL KICKS ²	KICK-OFFS ²	FREE KICKS ³	CORNER KICKS ³

¹ includes kick-outs from the goalkeeper's hands

² played as passes

³ played as crosses

⁴ without crosses from throw-ins

⁵ without passes from throw-ins

⁶ without passes from throw-outs

Figure 12: Overview of different types of balls played and the cumulative rules

Goal involvements						
Direct goal involvements		Indirect goal involvements				
Successful shots (goals)	Assists	2nd Assists	Other assists	Indirect assists	Free kick assists	Fouled player assists

Figure 13: Overview of different kinds of goal involvements and the cumulative rules

Shot involvements						
Direct shot involvements		Indirect shot involvements				
Shots	Direct shot setups	2nd Assists	Other shot setups	Indirect shot setups	Free kick shot setups	Fouled player shot setups

Figure 13a: Overview of different kinds of shot involvements and the cumulative rules



Figure 13b: Overview of different kinds of involvements in chances and clear-cut chances



*without throw-in crosses

Figure 13c: Overview of different kinds of balls into the area

4. POSITIONAL DATA

Figure 14 shows the basic structure of positional raw data. These are made up of *positional frames*, which contain information about the objects located on the pitch, recorded at a sampling rate of 25 Hz. Every positional frame aggregates frame data, personal data and ball data. *Frame data* contain time data as well as information on the number of present data packets. *Personal data* contain the person ID, position, speed, as well as distance and acceleration values for every person involved in the match, including the referees (usually 25 people in total). *Ball data* contain the position of the ball as well as information about the status of the ball (Ball in play, Ball not in play) and about possession (home team or away team). In addition to the frames, *positional metadata* (match ID, field size) are available for the interpretation of the positional frames.

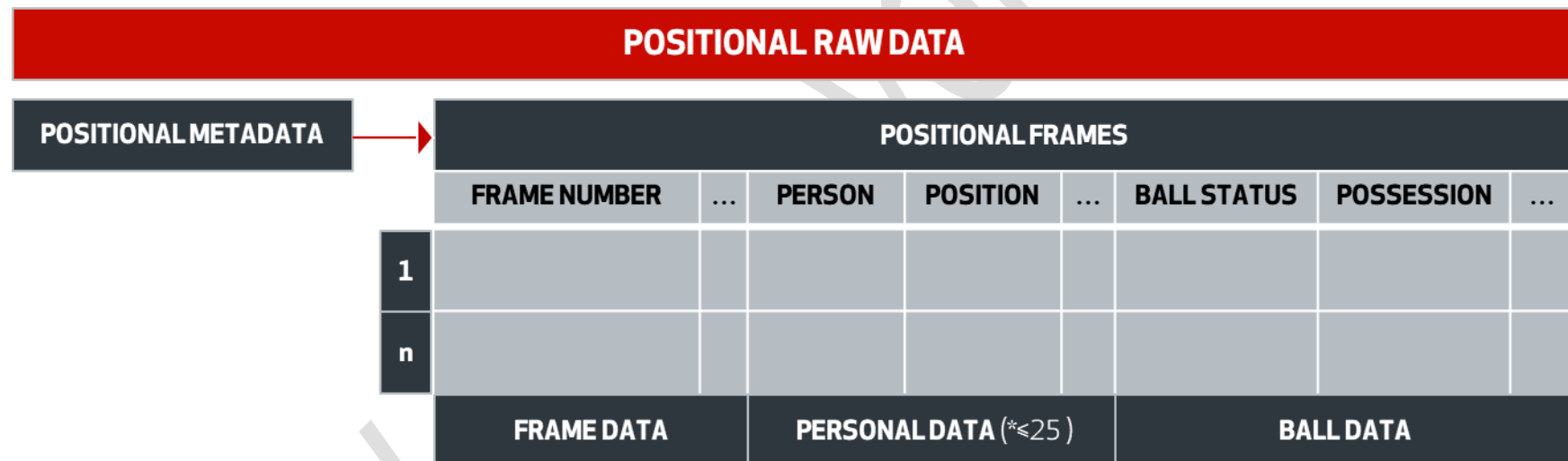


Figure 14: Structure of positional raw data (simplified)

Positional frames are available for the time between the start and end of the periods of play (indicated by the referee's whistle). The only exception is when the referee has halted the game and the match event other referee action is recorded (see attributes start of interruption of play and end of interruption of play). No positional data are available for penalty shootouts.

A detailed description of the technical data structure can be found under [2] of the respective version of the document DFL-04.01-Positionsdaten-Spiel-Roh, in the description of the respective services offered.

4.1 POSITIONAL METADATA

Description	Positional metadata are multi-frame data on the interpretation of the positional frames.	
Fields/allocations	<i>Pitch length x-axis</i>	[float]
	<i>Pitch length y-axis</i>	[float]
Additional description	<i>Pitch lengths</i> indicates the length of the touchlines (x-axis) and the goal lines (y-axis) in metres (two decimal places, rounding method “half up”)	

4.2 FRAME DATA

Description	Frame data are supplied at the start of each positional frame and contain metadata concerning personal data and ball data.	
Fields/allocations	Time stamp	[time stamp]
	Minute of play	[numeric]
	Frame number	[numeric]
	Period of play	First half Second half First half of extra-time Second half of extra-time

Additional description	Time stamp indicates the absolute point in time represented by the present data. Minute of play indicates the point in time represented by the present data relative to the start of the respective period of play. In the first half, the first minute of play refers to the time between the referee's whistle at the start of the half and the first minute. Zeitstempel beschreibt den absoluten Zeitpunkt, der durch die enthaltenen Daten repräsentiert wird. In the case of extra time, the minute of play is not given by adding +/- but as minute 46, 47, etc. However, the second half then begins with minute 46. The frames' periods of play are distinguished by using the field period of play. Frame number is a unique identifier for a frame. The count continues from frame to frame for the entire duration of the period of play. The initial frame number differs for each period of play First half: 10,000 Second half: 100,000 First half of extra-time: 200,000 Second half of extra-time: 250,000 Match underway indicates whether a game is underway, has not yet begun or has been stopped at the time the frame is recorded. Before the start of the game and in between two periods of play the manifestation no is used. Number of players returns the total count of player data contained in the frame. In addition to player data, data on the referee and the assistant referees are also supplied.
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4.3 PERSONAL DATA

Description	Personal data contain a person's positional data for the point in time supplied by the frame data.	
Fields/allocations	Person	Reference person ID
	x-coordinate	[float]
	y-coordinate	[float]
	Distance	[float]

	Speed	[float]
	Acceleration	[float]
Additional description	Personal data are supplied for all players and referees involved in the game, provided they are positioned within the confines of the extended pitch (pitch plus an additional 10 metres behind the goal lines and an additional 3 metres behind the touchlines). Usually this includes 22 players, the referee and the two assistant referees – if a player has been dismissed or (temporarily) injured, the count is reduced accordingly. The x- and y-coordinates indicate the position of the people on the pitch in metres. The position refers to their projected centre of gravity on the pitch. The origin of the coordinate system is located at the centre spot of the pitch, the axes run parallel to the touchlines (x-axis) and the goal lines (y-axis), respectively. The direction of the axes is defined by the image plane of the optical tracking system (positive quadrant top/right). Distance is given in centimetres (cm) and indicates the distance covered since the preceding frame. Speed is given in unsigned km/h. Speed refers to the modulus of the velocity vector in the xy plane. Acceleration is given in signed m/s². Acceleration refers to the modulus of the acceleration vector in the xy plane. Distance, speed and acceleration of a person are calculated by applying a series of filter mechanisms to the coordinates and are described exclusively through the use of the values given here. A use of other quantities (e.g. calculations using the Euclidean distance between two coordinates) is not intended for the official match data. All metric values are rounded to two decimal places. The “half-up” method is used for rounding.	

4.4 BALL DATA

Description	Ball data contain the position of the ball as well as information about the status of play and about possession.	
Fields/allocations	x-coordinate	[float]
	y-coordinate	[float]
	z-coordinate	[float]
	Distance	[float]

	Speed	[float]
	Acceleration	[float]
	Status of play	In play Not in play
	Possession	Home team Away team
Additional description	The x-, y- and z-coordinates indicate the position of the ball on the pitch in metres. The position of the ball refers its centre of gravity. The same coordinate system is used as for ball data as the coordinate system used for personal data. Distance is given in centimetres (cm) and indicates the distance covered since the preceding frame. Speed is given in unsigned km/h. Speed refers to the modulus of the velocity vector in the xy plane. Acceleration is given in signed m/s². Acceleration refers to the modulus of the acceleration vector in the xy plane. Distance, speed and acceleration of a person are calculated by applying a series of filter mechanisms to the coordinates and are described exclusively through the use of the values given here. A use of other quantities (e.g. calculations using the Euclidean distance between two coordinates) is not intended for the official match data. Status of play indicates whether the ball is currently in play or not. The ball is considered not to be in play if the referee has halted the game in accordance with the DFB football rules (e.g. after a foul) or the ball has left the pitch. Possession indicates which the team currently has possession of the ball. A team is considered to be in possession once a player of that team touches the ball with ball control for the first time after an opponent's phase of possession. Possession lasts until the ball is out of play or the other team touches the ball with ball control again. If the ball is not in play according to the attribute status of play, then the last team to have had possession before the ball went out of play is recorded for the attribute possession. The period of time that the ball is not in play is not included in the calculation of statistical base units (see section 4.5) and is not recorded as either team's time in possession.	

4.5 STATISTICAL BASE UNITS FROM POSITIONAL DATA

Statistical base units from positional data are calculated on the basis of the frames supplied for the game in question. For example, top speed of a player is calculated by returning the maximum value of all speed values for that player; a team's possession is calculated on the basis of the ball status and possession fields (see figure 15).

Working version

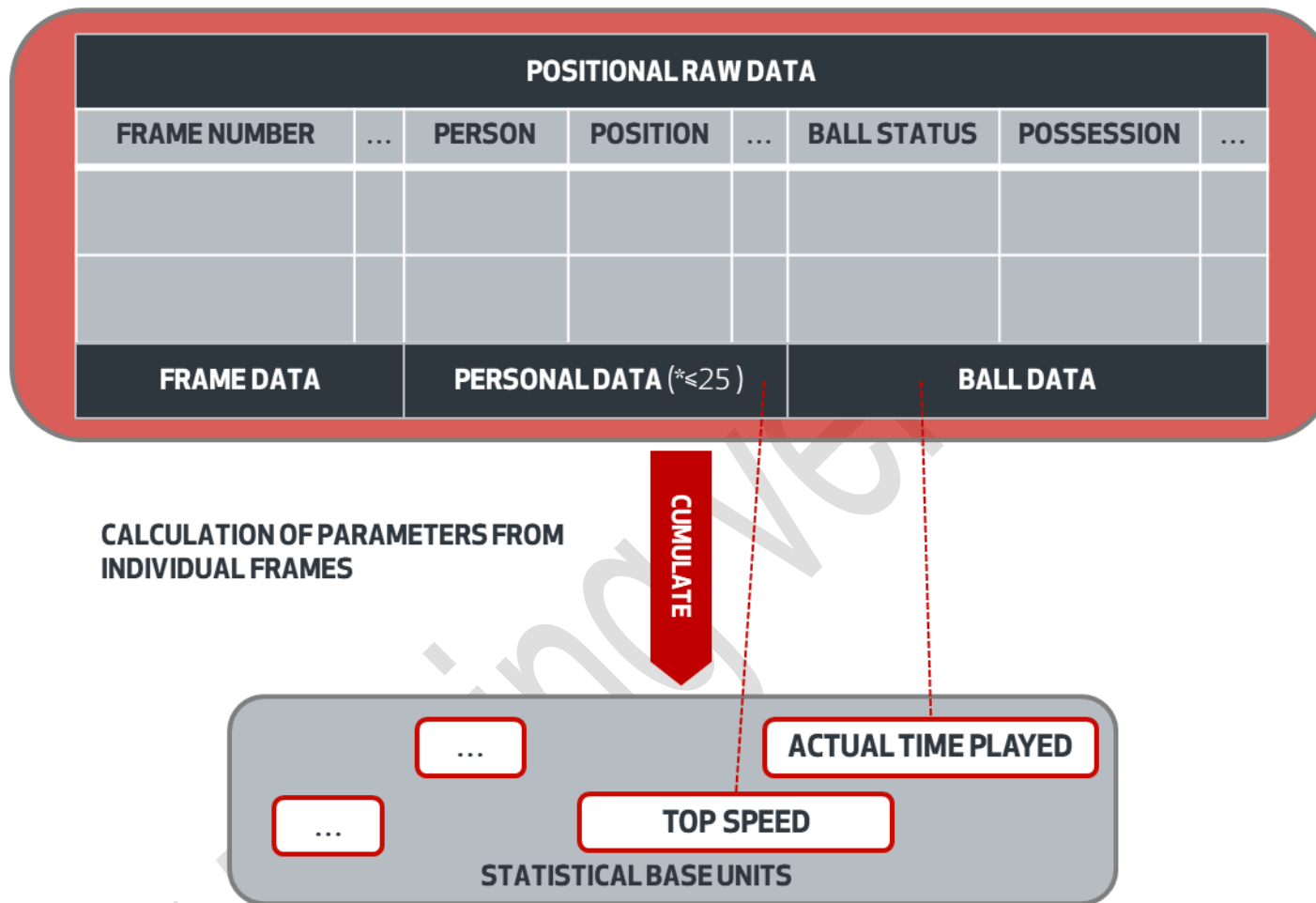


Figure 15: Calculation of statistical base units from positional raw data (simplified)

Table 2 describes how parameters such as sprints or distances covered in speed zones are calculated. A detailed description of the technical data structure can be found under [2]. The formalised rule can be found under [1]. When using an official statistical base unit, the exact official label defined here should be used.

Table 2: Statistical base units on the basis of positional raw data

<i>Official label</i>	<i>Description</i>
Distance covered	A player's distance covered between two frames refers to the distance value of the second frame.
Total distance covered	A player's (or team's or other entity's) total distance covered is calculated by adding up all distance values of the individual frames.
Average speed	A player's average speed is calculated by dividing his total distance covered by his total time played.
Top speed	A player's top speed refers to the maximum value of all speed values for that player.
Maximum acceleration	A player's maximum acceleration refers to the maximum value of all acceleration values for that player.
Maximum delay	A player's maximum delay refers to the minimum value of all acceleration values for that player.
Sprints	A sprint is defined as a player running at least 4.0 m/s for more than two seconds, during which he in turn achieves a speed of at least 6.3 m/s and the time between reaching the 6.3 m/s threshold for the first and last time is at least one second. The sprint time is limited to the period between reaching the 6.3 m/s threshold for the first and last time. A new sprint can only be recorded after the player has fallen short of the 4.0 m/s threshold. [1]
High-speed runs	A high-speed run is defined as a player running at least 4.0 m/s for more than two seconds, during which he in turn achieves a speed of at least 5.0 m/s and the time between reaching the 5.0 m/s threshold for the first and last time is at least one second. At the same time, the definition of a sprint must not be fulfilled. The high-speed run time is limited to the period between reaching the 5.0 m/s threshold for the first and last time. A new high-speed run can only be recorded after the player has fallen short of the 4.0 m/s threshold. [1]
Intensive runs	The total count of intensive runs refers to the sum of the total counts of all high-speed runs and sprints.
Distance covered with sprints	Represents the distance the player covered with sprints.
Distance covered with high-speed runs	Represents the distance the player covered with high-speed runs.

<i>Official label</i>	<i>Description</i>
Direction of movement	A player's direction of movement in a time interval is calculated from his location coordinates at the start and end of the interval. Forward: the x-coordinate is closer to the opponents' goal at the end of the interval than it was at the start of the interval Backward: in all other cases
Speed zones	1: <11 km/h 2: :11 - 14 km/h 3: 14 - 17 km/h 4: 17 - 21 km/h 5: 21 - 24 km/h 6: >24 km/h
Time in speed zone n	A player's total count of all frames for which the speed is within speed zone n, multiplied by the sampling time (reciprocal of the sampling rate; currently 40ms).
Distance covered in speed zone n	Represents the sum of all distances covered by a player between two consecutive frames, whereby the speed of the second frame is within the speed zone n.
Actual time played	The total count of all frames during which the ball is in play, multiplied by the sampling time (reciprocal of the sampling rate; currently 40ms).
Possession	The total count of all frames during which the ball is in play and the team in question is in possession, multiplied by the sampling time (reciprocal of the sampling rate; currently 40ms).
Percentage of possession	The ratio between the home and away team in regards to the percentage of possession (e.g. 55% : 45%).

Notes on the characteristics and processing of positional data

The recording of positional data is based on a semi-automated, optical multi-camera tracking system. Continuously determining an individual's position is a technically complicated process. Optical tracking systems must be able to recognize the players' shapes and distinguish between players under changing conditions. As a result, the positional data should reflect changes in the direction of movement and the body's centre of gravity in detail. On the other hand, the trajectory should also match the perceived, smooth movement path.

In order to ensure this, filter mechanisms are implemented in tracking systems, which generate coherent positions, distances, speeds and acceleration values in several processing steps. To give an impression of the sensitivity and smoothing of the raw positional data supplied by the DFL, figure 16 shows the movements of a player over a period of 500 frames (20 s). Figure 17 shows an enlargement of the change of direction over a period of 135 frames (5.4 s).

The filter mechanisms of the tracking system are always discussed in detail and disclosed to the DFL, to ensure the highest possible live data recording rate.

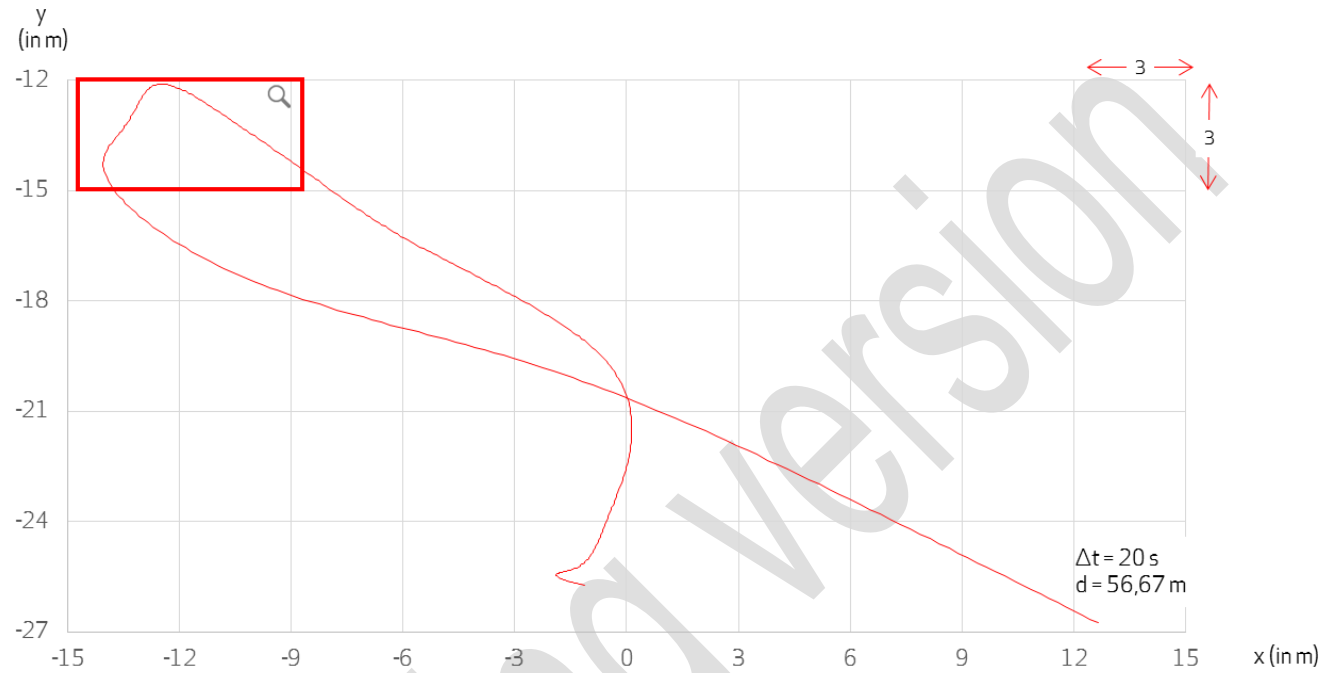


Figure 16: Path of an outfield player over a period of 20 seconds. The distance covered in this time is 56.67 metres

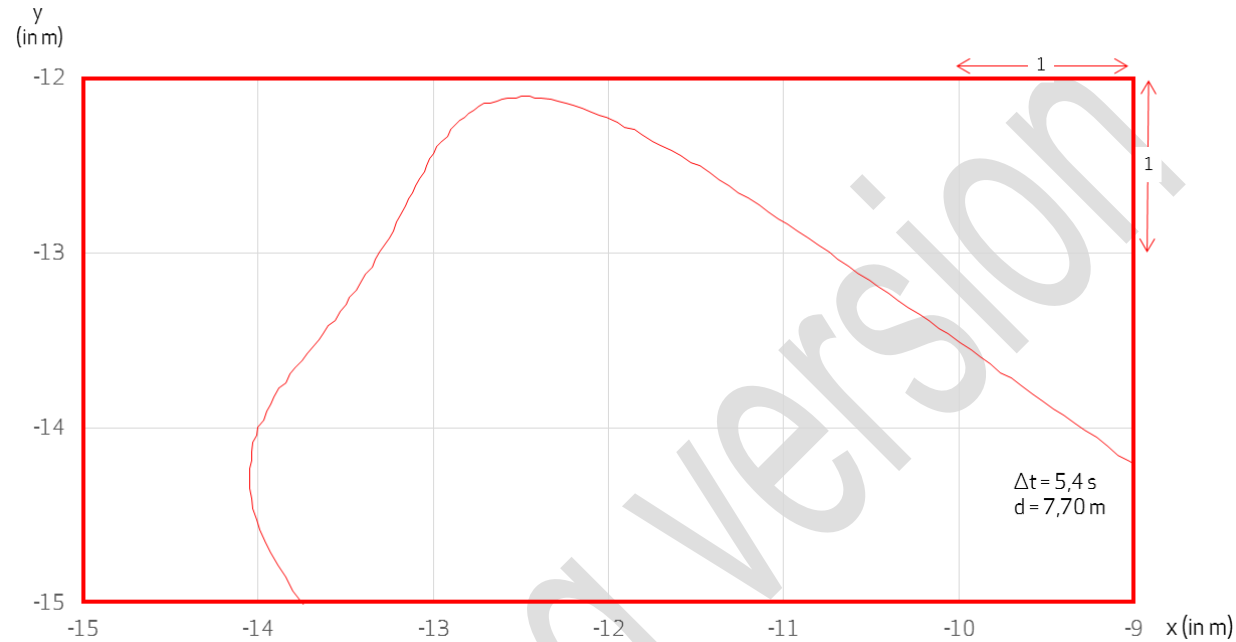


Figure 17: Enlarged depiction of the change of direction shown in figure 16 (5.4s); the distance covered in this section is 7.7 metres

REFERENCES

- [1] Deutsche Fußball Liga: Formulary regarding the calculation of statistics from official match data, v3.1 June 2015.
- [2] Deutsche Fußball Liga: DFL-04.01–Positionsdaten-Spiel-Roh

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